
Learned spectral projections in place of the bispectrum: a compact descriptor for the quadratic SNAP energy form

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Abstract

Quadratic SNAP buys accuracy by enriching the regression around a fixed descriptor: adding all pairwise products of the 55 bispectrum components cuts the energy error on tantalum by a factor of three, at the cost of raising the number of fitted coefficients from 55 to 1595 and of a cross-validation curve that diverges when less than about a third of the training set is used. We run the complementary experiment — hold the regression fixed and learn the descriptor — and report what it costs and what it buys. CLASP (Compact Learned Atomic Spectral Projections) expands the neighbour density on a learned radial basis, contracts the resulting equivariant moments into second- and third-order rotation invariants with the Gaunt coupling tensor, and projects them onto K' learned coordinates that feed an otherwise unchanged quadratic SNAP model: same energy decomposition, same zero-isolated-atom constraint, same reference potential, same analytic forces, same least-squares solver. Rotation, translation and permutation invariance are exact by construction; audited against a rebuilt periodic neighbour list under 1000 random rotations the measured drift is 10^{-8} in relative energy, which we show is floating-point summation order amplified by the fitted coefficients rather than an approximation in the architecture. On the 363-configuration tantalum set of Wood and Thompson at their published protocol, CLASP with $K' = 16$ — 152 regression coefficients instead of 1595 — is more accurate in energy and statistically indistinguishable in force at full data, and over the paper’s 5%-increment, ten-draw cross-validation sweep the two descriptors differ by a regime rather than by a constant: below about a third of the data the learned descriptor is two to three orders of magnitude more accurate out of sample, and what it achieves with 5% of the configurations takes quadratic SNAP 60–90%, while above 40% the fixed descriptor is marginally ahead. A rank-reduced quadratic-SNAP control that shares the coefficient count but compresses the bispectrum linearly instead of learning it is an order of magnitude worse, which locates the gain in the learned nonlinearity rather than in the smaller regression. A head-matched variant that reads the same equivariant moments in the laboratory frame is, on this benchmark’s own metrics, *competitive* — better in force at full data than either the invariant descriptor or quadratic SNAP — while drifting by 1.3×10^{-2} in energy and $2.1 \text{ eV/\AA}$ in force under rotation, so it is not a potential energy surface at all. Held-out error cannot tell the two apart, because held-out configurations share the training set’s orientation distribution; only the audit can. Training that variant with random rotations reduces its drift threefold and costs a factor of five in accuracy. The same architecture, unchanged, transfers to molybdenum, niobium and hexagonal rhenium under the same protocol; on those sets, an order of magnitude larger and never in the baseline’s data-starved regime, the learned descriptor is the more accurate of the two at full data on two of the three and clearly worse on the third, and we say why.

1 Introduction

Machine-learned interatomic potentials in the density-projection family — SNAP, the atomic cluster expansion, moment tensor potentials — factor the map from an atomic neighbourhood to a per-atom energy into two stages. A *descriptor* compresses the neighbourhood $\{\mathbf{r}_{ij}\}$ into a fixed-length vector $\mathbf{B}^i \in \mathbb{R}^K$ that is invariant to rotation, translation and permutation of the neighbours by construction; a *regression model* then maps $\mathbf{B}^i$ to E^i . Wood and Thompson [20] showed that keeping the descriptor fixed and enriching only the regression — from a linear function of the $K = 55$ bispectrum components to a linear-plus-quadratic one — lowers the mean absolute energy error on a tantalum training set by a factor of 3.4, from 0.03041 to 0.00898 eV/atom, at less than twice the cost per timestep. Their analysis also exposed the price: the quadratic form has $K + K(K+1)/2 = 1595$ fitted coefficients instead of 55, and its cross-validation curves diverge catastrophically below roughly 30% of the training set.

That result invites the complementary experiment. If enriching the regression around a fixed descriptor buys accuracy at the cost of data, what happens if the *descriptor* is learned instead, with the regression held fixed at the linear-plus-quadratic form? The bispectrum is a complete but generic basis: it is derived from the symmetry of the 3-sphere, not from the material, and its size grows steeply with angular resolution ($K = 55$ at $J_{\max} = 4$, 91 at 5). A learned descriptor can in principle place its capacity where the data demands it — around the interatomic distances and bond angles the material actually samples — and can do so in far fewer dimensions. Because the number of fitted coefficients grows quadratically in the descriptor dimension, a descriptor that is even modestly more compact translates into a much smaller regression problem, which is precisely the quantity that governs the overfitting Wood and Thompson documented. The counterweight is that the three invariances are no longer free: they must be built into the learned architecture, learned from data, or supplied by augmentation, and each choice has consequences for sample efficiency and for behaviour outside the training distribution.

This paper carries out that experiment. We introduce CLASP (Compact Learned Atomic Spectral Projections), a learned descriptor that replaces the bispectrum inside an otherwise unchanged quadratic SNAP model: same energy decomposition $E^i = \beta \cdot \mathbf{B}_\theta^i + \frac{1}{2}(\mathbf{B}_\theta^i)^\top \alpha \mathbf{B}_\theta^i$, same zero-isolated-atom constraint, same ZBL reference potential, same analytic forces, and (β, α) still obtained by the same truncated-SVD least-squares solve. CLASP expands the neighbour density on a *learned* radial basis and real spherical harmonics, contracts the resulting equivariant moments into rotation invariants of order two and three with the Gaunt coupling tensor, and projects those invariants onto $K' = 16$ learned coordinates. The invariant contraction makes rotation, translation and permutation invariance exact, and the projection makes the regression problem $10.5\times$ smaller than quadratic SNAP’s.

Our contributions and findings are as follows.

(i) A learned descriptor that is smaller and no less accurate. On the tantalum set of Wood and Thompson [20] at the published protocol, CLASP with $K' = 16$ reaches full-data errors below our reimplemented quadratic SNAP in both energy and force, with 152 instead of 1595 regression coefficients (Table 2).

(ii) A data-efficiency result that is reported where it is discriminating. Over the paper’s full 5%-increment, ten-draw sweep, CLASP reaches quadratic SNAP’s full-data energy error at a small fraction of the data, and the gap is largest exactly in the regime Wood and Thompson identified as the failure mode of the quadratic form. We report the outcome separately for the equal-weight aggregate over all twelve training groups and for the low-symmetry subset (GSF, liquid, surface), because the high-symmetry groups saturate at tiny fractions for both descriptors and cannot discriminate between them.

(iii) Evidence that the cross-validation protocol cannot see a broken symmetry. Swapping only the contraction stage — Gaunt-contracted invariants versus the same moments read out in the laboratory frame — gives a non-invariant descriptor with the same dimension and the same head. Run through the same sweep it is *competitive*: better in force at full data than either the invariant descriptor or quadratic SNAP, and better in the all-group held-out median at every fraction we measured. It is also not a potential energy surface: rotating a configuration moves its energy by 1.3% and its forces by more than the typical force in the dataset. Held-out error cannot distinguish the two, because held-out configurations share the training set’s orientation distribution; only the invariance audit can.

Rotation augmentation reduces the drift threefold and costs a factor of five in accuracy, which is the price of learning a symmetry instead of encoding it.

(iv) Transfer to three further refractory metals. The same architecture, retrained per material, is evaluated on Mo, Nb and Re from the single-element deep-potential collection of Andolina and Saidi [1], with Re’s HCP symmetry acting as the structural-generalisation test. We report both the cases where the learned descriptor wins and the cases where it does not.

2 The quadratic SNAP form and what the descriptor does in it

SNAP writes the total energy of a configuration as a short-ranged reference potential plus a sum of per-atom contributions, $E(\mathbf{r}^N) = E_{\text{ref}}(\mathbf{r}^N) + \sum_i E^i$, and expands the neighbour density of atom i , $\rho_i(\mathbf{r}) = \delta(\mathbf{r}) + \sum_{\mathbf{r}_{ij} < R_{\text{cut}}} f_c(r_{ij}) w_j \delta(\mathbf{r} - \mathbf{r}_{ij})$, in four-dimensional hyperspherical harmonics after mapping the radial coordinate onto a third angle [2, 18]. The rotation-invariant cubic contractions of the expansion coefficients are the bispectrum components $B_{j_1 j_2 j}$, and their number K is set by the truncation J_{max} . Quadratic SNAP [20] takes

$$E_{\text{SNAP}}^i = \boldsymbol{\beta} \cdot \mathbf{B}^i + \frac{1}{2} (\mathbf{B}^i)^\top \boldsymbol{\alpha} \mathbf{B}^i, \quad (1)$$

with $\boldsymbol{\alpha}$ symmetric. Equation (1) is the second-order Taylor expansion of an embedding energy $F(\rho_i)$ in a density $\rho_i = \mathbf{a} \cdot \mathbf{B}^i$ that is itself linear in the bispectrum, which is what makes the form EAM-like; because the energy is still linear in $(\boldsymbol{\beta}, \boldsymbol{\alpha})$, the fit remains a linear least-squares problem, and the forces $\mathbf{F}^j = -\nabla_j E$ are exact derivatives of the descriptor.

Two structural facts about Eq. (1) organise everything that follows.

First, the descriptor enters only through the $P = K + K(K+1)/2$ scalar features

$$\left[\sum_i B_f^i \right]_{f \leq K} \quad \text{and} \quad \left[\sum_i \frac{1}{2} (B_f^i)^2 \right]_f \cup \left[\sum_i B_f^i B_g^i \right]_{f < g}, \quad (2)$$

which is the design matrix of the least-squares fit — its force rows are the negative coordinate derivatives of the same quantities. Anything that supplies a per-atom vector $\mathbf{B}^i$ and its analytic Jacobian therefore drops into the SNAP machinery unchanged. This is the interface CLASP plugs into.

Second, P is quadratic in the descriptor dimension. At $J_{\text{max}} = 4$, $K = 55$ gives $P = 1595$ against 363 energy rows and 12,672 force rows in the tantalum set: a factor of eight more rows than coefficients at full data, but fewer rows than coefficients below about 12% of the data, and empirically the prediction error is still dominated by the ill-conditioning of the fit well beyond that. Halving K divides P by nearly four. The descriptor dimension is thus not just a cost knob: it is the main lever on the data requirement of the quadratic form. Any claim that a learned descriptor is more data-efficient has to be tested against the possibility that a smaller *fixed* descriptor would do just as well, which is why we carry a rank-reduced quadratic-SNAP control (Sec. 4) alongside the two required baselines.

A third fact is about invariance rather than dimension. The bispectrum is invariant by construction: rotating a configuration leaves $\mathbf{B}^i$ unchanged to machine precision and rotates the forces exactly, whatever the training data contained. Every configuration therefore teaches the model only about the physics, never about the orientation, and this is usually credited for the sample efficiency of the SNAP family. A learned descriptor forfeits that guarantee unless the architecture restores it. We quantify what it is worth in Sec. 6.

3 CLASP: compact learned atomic spectral projections

CLASP maps the neighbourhood of atom i — the set of displacement vectors $\mathbf{d}_{ij} = \mathbf{r}_j + \mathbf{s}_{ij} - \mathbf{r}_i$ to all periodic images inside r_{cut} — to a vector $\mathbf{B}_\theta^i \in \mathbb{R}^{K'}$ in four stages. Everything in this section is fixed before any result in Sec. 5 is quoted; the hyper-parameter values are collected in Table 1 and are the ones written into the run logs.

1. Learned radial encoder. Each neighbour distance is expanded on the sine (Bessel-type) basis $\phi_n(r) = \sqrt{2/r_{\text{cut}}} \sin(n\pi r/r_{\text{cut}})/r$, $n = 1 \dots N_b$, and passed through a two-hidden-layer multilayer perceptron with SiLU activations, giving C radial channels multiplied by a smooth envelope,

$$g_c(r) = u(r) \text{MLP}_c(\phi_1(r), \dots, \phi_{N_b}(r)), \quad u(r) = 1 - 28x^6 + 48x^7 - 21x^8, \quad x = r/r_{\text{cut}}, \quad (3)$$

with $u(r_{\text{cut}}) = u'(r_{\text{cut}}) = u''(r_{\text{cut}}) = 0$ so that energies and forces are twice differentiable at the cutoff. The nonlinearity matters: a *linear* map of a fixed basis only re-coordinatises its span and could be absorbed downstream, whereas the MLP can synthesise radial filters sharper than any ϕ_n — shell-selective profiles tuned to the material’s radial distribution — which is where the descriptor’s material specificity lives.

2. Equivariant moments. With $\hat{\mathbf{n}}_{ij} = \mathbf{d}_{ij}/r_{ij}$ and real spherical harmonics $Y_{\ell m}$ up to $\ell \leq L$,

$$A_{c\ell m}^i = \sum_{j \neq i} g_c(r_{ij}) Y_{\ell m}(\hat{\mathbf{n}}_{ij}). \quad (4)$$

These are exactly translation invariant (they see only displacements), permutation invariant (a sum over neighbours) and rotation *equivariant*: each ℓ -block transforms under the Wigner matrix $D^{(\ell)}$.

3. Invariant contraction. Two blocks of invariants are formed from Eq. (4). The $\nu = 2$ block is the full cross-channel power spectrum

$$p_{cc'\ell}^i = \sum_m A_{c\ell m}^i A_{c'\ell m}^i, \quad c \leq c', \quad (5)$$

which is invariant because it contracts two equivariant vectors with the $\text{SO}(3)$ -invariant metric. The $\nu = 3$ block contracts three moments with the real Gaunt tensor $G_{m_1 m_2 m_3}^{\ell_1 \ell_2 \ell_3} = \int Y_{\ell_1 m_1} Y_{\ell_2 m_2} Y_{\ell_3 m_3} d\Omega$, itself an invariant tensor of $\text{SO}(3)$:

$$b_{s,(\ell_1 \ell_2 \ell_3)}^i = \sum_{m_1 m_2 m_3} G_{m_1 m_2 m_3}^{\ell_1 \ell_2 \ell_3} \tilde{A}_{s\ell_1 m_1}^{i,1} \tilde{A}_{s\ell_2 m_2}^{i,2} \tilde{A}_{s\ell_3 m_3}^{i,3}, \quad \tilde{A}_{s\ell m}^{i,q} = \sum_c V_{cs}^q A_{c\ell m}^i, \quad (6)$$

over the $(\ell_1 \leq \ell_2 \leq \ell_3)$ triples that satisfy the triangle rule with even $\ell_1 + \ell_2 + \ell_3$ (14 triples at $L = 4$). The $\nu = 3$ block is what makes CLASP four-body in the same sense as the bispectrum, so that the quadratic head reaches the same seven-body complexity as quadratic SNAP; an ablation with the block removed is reported in Sec. 6. The three slot-mixing matrices V^1, V^2, V^3 per triple are *learned*: SNAP couples fixed hyperspherical orders with fixed Clebsch–Gordan weights, whereas here which radial shells are correlated at third order is fitted. The Gaunt tensors are precomputed by Gauss–Legendre $\times$ uniform- φ quadrature, exact for the polynomial integrands involved.

4. Learned projection and the zero-energy constraint. The concatenated invariants $\mathbf{z}^i$ are compressed and standardised, then projected,

$$\mathbf{B}_\theta^i = \text{MLP}_{\text{out}}(\sigma(\mathbf{z}^i)) - \text{MLP}_{\text{out}}(\sigma(\mathbf{0})), \quad \sigma(\mathbf{z}) = \frac{\text{arcsinh}(\mathbf{z}/\mathbf{s}) - \boldsymbol{\mu}}{\boldsymbol{\varsigma}}. \quad (7)$$

The $\nu = 2$ and $\nu = 3$ invariants are second and third powers of the density amplitude, so their raw distributions are heavy-tailed — a compressed volumetric cell and a free surface differ by orders of magnitude — and arcsinh is a smooth, odd, monotone map that equalises the dynamic range while preserving the exact zero at $\mathbf{z} = \mathbf{0}$. The statistics $(\mathbf{s}, \boldsymbol{\mu}, \boldsymbol{\varsigma})$ are computed from the included configurations of the current fit only. Subtracting the isolated-atom value in Eq. (7) is the learned analogue of SNAP’s B_{k0}^i shift: an atom with no neighbours has $A_{c\ell m}^i = 0$, hence $\mathbf{B}_\theta^i = \mathbf{0}$ and $E^i = 0$ exactly, which is the `bzeroflag=1` condition.

Invariance by construction, and the head-matched control. Stages 1, 2 and 4 are invariant for any descriptor built on displacement sums; stage 3 is what removes orientation. We therefore define a second variant, **CLASP-C**, in which stage 3 is replaced by flattening the same moments $A_{c\ell m}^i$ in the laboratory frame. CLASP-C has the same radial encoder, the same moments, the same projection-MLP widths, the same descriptor dimension K' and therefore the same 152-coefficient head, and the same training procedure, and it sees strictly *more* information (orientation included); it differs from CLASP only in whether that orientation is contracted away. The one quantity it does not match is the descriptor parameter count: flattening gives $C(L+1)^2 = 400$ inputs to the projection MLP where the contraction gives 904, and the slot-mixing matrices disappear with the $\nu = 3$ block, so CLASP-C carries 185,312 parameters against CLASP’s 314,336. It is therefore matched on the variable this study holds fixed — the regression — and is given strictly more information from a slightly smaller network. CLASP-C is run both without augmentation and with random global rotations applied to every training configuration at every step. This triple — exact, augmented, neither — is the controlled experiment behind the invariance discussion.

Table 1: CLASP hyper-parameters. These values are fixed before any result is quoted and are identical for all four materials and for every cell of every sweep; the only per-material quantity is the cutoff, which is inherited from the quadratic-SNAP baseline (Sec. 4). The values are the ones written into every run log (proposal/logs/sweeps/*.log).

Stage	Quantity	Value
Radial encoder	cutoff r_{cut} (Å)	5.594 (Ta), 5.3 (Mo, Nb), 5.0 (Re)
	sine basis functions N_b	12
	MLP widths (SiLU)	12 $\rightarrow$ 32 $\rightarrow$ 32 $\rightarrow$ 16
	envelope exponent p	6
Moments and invariants	radial channels C	16
	angular truncation L	4
	$\nu=2$ invariants $C(C+1)/2 \times (L+1)$	680
	$\nu=3$ triples $\times$ slots 14×16	224
Projection	MLP widths (SiLU)	904 $\rightarrow$ 256 $\rightarrow$ 256 $\rightarrow$ 16
	descriptor dimension K'	16
	regression coefficients $P = K' + K'(K'+1)/2$	152
Training	optimiser	Adam, one-cycle schedule
	peak learning rate	3×10^{-3} ($\times 10$ for the temporary head)
	weight decay / gradient clip	10^{-6} / 10
	steps at included fraction f	6000 + 24000 f
	configurations per step	16
	collapse threshold / max restarts	0.003 / 2
	precision	float64 throughout
	descriptor parameters θ	314,336

Energy model, forces and the fit. The per-atom energy is Eq. (1) with $\mathbf{B}^i$ replaced by $\mathbf{B}_\theta^i$; forces are exact analytic gradients obtained by differentiating the descriptor. For the least-squares fit we assemble the same $P = K' + K'(K'+1)/2$ features of Eq. (2) and their coordinate derivatives. The per-atom descriptor Jacobian $\partial \mathbf{B}^i / \partial \mathbf{d}_m^i$ is obtained by one reverse-mode pass batched over the K' descriptor components and mapped onto atomic coordinates through $\mathbf{d}_m^i = \mathbf{r}_{n(i,m)} + \mathbf{s} - \mathbf{r}_i$; this is exact, and we verify it against both finite differences and direct autodifferentiation of the total energy (Appendix A.1). The coefficients (β, α) are then obtained with the same solver FitSNAP uses — the minimum-norm truncated-SVD solution with singular values below $10^{-13} s_{\text{max}}$ discarded — on the included portion only.

Training the descriptor. The descriptor parameters θ are trained by stochastic gradient descent on minibatches of included configurations, jointly with a temporary free (β, α) , minimising $\mathcal{L} = \langle (\Delta E / \varsigma_E)^2 \rangle + \langle (\Delta F / \varsigma_F)^2 \rangle$ with ς_E, ς_F the standard deviations of the included targets; θ is then frozen and the head is discarded and refitted by exact least squares as above. Adam with a one-cycle schedule is used, with minibatches of 16 configurations and a step budget that scales with the amount of data being fitted, $\text{steps} = 6000 + 24000f$ at included fraction f ; the rule is fixed in advance and applied identically at every cell, material and variant.

Stochastic training of a descriptor has a failure mode that a linear least-squares fit does not: in a fraction of runs $\mathbf{B}_\theta^i$ collapses to a constant, after which the head can only produce a constant per-atom energy and zero forces. We detect this from the variance-normalised residual of the exact least-squares fit *on the included data* — near unity for a collapsed descriptor, three orders of magnitude below it for a healthy one — and restart from a new initialisation, at most twice, keeping the last attempt if none passes. The criterion uses training data only, and the number of restarts used is written to every per-cell log.

Crucially, the descriptor is retrained from scratch for *every* cell of the cross-validation sweep, using only that cell’s included configurations: nothing about the held-out portion — not the standardisation statistics, not the minibatch sampling, not the collapse criterion, not the stopping point — enters training. A second training procedure that differentiates through a closed-form least-squares head on random support/query splits (“solver-in-the-loop”) is reported as an ablation in Sec. 6; a third, alternating variable projection, was tried and set aside (proposal/attempts_log.md).

4 Experimental design

Data. The tantalum experiments use the 363 DFT configurations distributed with FitSNAP for Wood and Thompson [20], organised in the twelve groups `Displaced_{A15,BCC,FCC}`, `Volume_{A15,BCC,FCC}`, `Elastic_{BCC,FCC}`, `GSF_{110,112}`, `Liquid` and `Surface`. No configuration was added, removed or synthesised. We note a provenance discrepancy that matters for reading the two-baseline comparison: the distributed set contains 363 energies and 12,672 force components, whereas Wood and Thompson [20] describe 362 energies and 15,297 force components, with the difference concentrated in the `Liquid` and `Elastic` groups. We fit what is distributed and flag the gap in Sec. 5.

The transfer materials Mo, Nb and Re are the pre-staged single-element deep-potential datasets of Andolina and Saidi [1] (GPL-3.0), taken from the `iter*.tar.xz` archives verbatim: 3727 Mo, 3246 Nb and 5420 Re configurations, i.e. 203,923, 172,404 and 341,822 design rows respectively. They are adaptive-learning-refined molecular-dynamics trajectories at several temperatures with elastic, vacancy and interstitial perturbations. We convert them from DeePMD to FitSNAP JSON and form configuration groups as `category_prototype_T` from the upstream directory layout (`category` $\in$ {`elastic`, `vacancy`, `interstitial`, `MD`}), giving 38, 40 and 20 groups — the closest analogue of tantalum’s thermodynamic grouping that the source data supports.

Two quadratic-SNAP baselines. Every comparison carries both baselines required to interpret it. *Paper-as-reported* quadratic SNAP is the published value $\delta E = 0.00898$ eV/atom at $J_{\max} = 4$, $r_{\text{cut}} = 5.594$ Å (Table SII of Wood and Thompson [20]); the linear counterpart is 0.03041 eV/atom at $r_{\text{cut}} = 5.672$ Å (Table SI). *Reimplemented* quadratic SNAP is our own FitSNAP run on the same 363 configurations at the same hyper-parameters (`twojmax=8`, `rcutfac=5.594`, `rfac0=0.99363`, `bzeroflag=1`, `quadraticflag=1`, ZBL reference `zbl 4.0 4.8`) and is the object every paired comparison is made against. Our FitSNAP/LAMMPS build reproduces the reference metrics shipped with the example deck to all printed digits (`proposal/logs/fitsnap_reference_check.md`).

Fitting protocol. All fits — quadratic SNAP, the rank-reduced control and CLASP — use the *unweighted* least-squares setting, $\gamma_n = 0$ and $\mathbf{w} = \mathbf{I}$, which is the setting Wood and Thompson [20] specify for the cutoff and accuracy tables (their Sec. V) and the one under which our reimplemention reproduces Table SII most closely. Energy rows are per-atom normalised and force rows are raw components, exactly as FitSNAP assembles them; virial rows are excluded. We do not run the DAKOTA genetic-algorithm search over group weights that the source paper uses for its production potentials: introducing a stochastic outer loop over weights would make both arms of the comparison depend on it and would break the anchor to Table SII. This is a disclosed deviation, and we report a weighted-protocol robustness check in Appendix A.7.

Radial cutoff. For tantalum we adopt the published optimum $r_{\text{cut}} = 5.594$ Å for *both* methods, so that CLASP and quadratic SNAP see identical neighbourhoods and neither is tuned against the other. For the transfer materials no published optimum exists, so we repeat the source paper’s procedure — scan r_{cut} , fit unweighted least squares on the full data, take the minimiser of the training energy error — for quadratic SNAP, and hand the resulting cutoff to CLASP unchanged. The scans (Appendix A.2) give 5.3 Å for Mo and Nb and 5.0 Å for Re. Because the cutoff is selected on full-data training error for the baseline and inherited by the learned descriptor, the selection is if anything conservative for CLASP.

Cross-validation. We reproduce the Fig. 5 protocol of Wood and Thompson [20]: included fractions 5%, 10%, ..., 95% plus the full-data point, ten random draws per fraction, stratified within every configuration group with FitSNAP’s own rule $n_{\text{inc}} = \max(1, \lfloor fn + \frac{1}{2} \rfloor)$, and a configuration always enters or leaves as a whole. The split tensor is generated once, seeded on the fraction *value* so that it is independent of how the sweep is sharded across processes, and is consumed by every method, making all comparisons paired draw-by-draw. We report mean $\pm$ standard deviation over the ten draws for both δE (eV/atom) and δF (eV/Å) on the held-out and the included portions, for every group and for two aggregates: `*ALL`, pooled over rows, and `*EQGROUP`, the unweighted mean of the per-group errors. The low-symmetry subset `*LOWSYM` is the same equal-weight mean restricted to `GSF_110`, `GSF_112`, `Liquid` and `Surface`. Every (material, method, fraction, draw, group, split) cell is written to `proposal/logs/cv/`.

We define the sample-efficiency statistic F^* as the smallest included fraction at which a method’s mean held-out δE first drops to the *reimplemented* quadratic-SNAP full-data δE and stays at or below it for all larger fractions. The reference is the same full-data budget for every material and every method, and F^* is reported separately for *EQGROUP and *LOWSYM.

Controls and ablations. Besides linear SNAP, we run a *rank-reduced* quadratic SNAP: the descriptor $\mathbf{B}^i \in \mathbb{R}^{55}$ is replaced by $W\mathbf{B}^i \in \mathbb{R}^k$ with W the top- k right singular vectors of the included-portion descriptor matrix (uncentred, so $\mathbf{B} = 0 \mapsto 0$ and the isolated-atom energy stays zero). Because a linear compression of $\mathbf{B}$ maps the quadratic columns of the SNAP design matrix onto the compressed ones exactly, this control shares the baseline’s descriptors, targets and force rows and isolates *dimension* from *learning*. Ablations of CLASP cover the descriptor width K' , the $\nu=3$ block, the invariance mechanism (CLASP-C with and without rotation augmentation) and the training procedure. The two lab-frame variants are the one place where the fraction grid is reduced: they are swept at $\{5, 10, 20, 40, 60, 100\}\%$ with the full ten draws each, rather than at all nineteen fractions, because they enter the paper as a mechanism control and not as a candidate potential. The four headline cross-validation sweeps — tantalum and the three transfer materials — are run at the full protocol, and it is complete: all $4 \times (19 \times 10 + 1) = 764$ cells of the learned-descriptor sweep, and every cell of all four SNAP baselines, were run (Appendix A.4). The sweep was executed in two stages — a sharded first pass, then dedicated workers for the cells the first pass had not finished — because each learned-descriptor cell retrains a descriptor from scratch. The data split of a cell is reproducible from the fraction value alone, but the descriptor initialisation seed is derived from the cell’s position in the sweep, so a repeat run of an already-completed cell would be a different initialisation on the same split; the analysis therefore de-duplicates on (material, method, fraction, draw, split, group) with a fixed precedence that keeps the first pass’s run wherever one exists, so the later workers only ever filled cells that were empty and never replaced a completed one.

Apart from the two lab-frame variants, the ablations are reported at the full-data point only (Appendix A.3), together with the invariance audit, because the per-cell retraining that the protocol demands makes a full sweep per ablation roughly as expensive as a headline material and the compute did not stretch to one for each. This is a compute-driven reduction, it applies only to the ablations, and it means the evidence behind the width, $\nu=3$ and training-procedure ablations in Sec. 6 is full-data accuracy rather than data-efficiency curves.

Invariance audit. On a held-out portion of a fixed draw (80% included, draw 0) we take ten random held-out configurations and apply 100 uniformly random $\text{SO}(3)$ rotations, 10 translations with each component drawn from $\mathcal{U}[-1, 1] \text{ \AA}$, and 10 random relabellings of the atoms. Every transformed configuration is re-wrapped and its periodic neighbour list is rebuilt from scratch, so the test exercises the model end to end rather than an algebraic identity on a cached list. We report $\delta E_{\text{rel}} = |E' - E|/|E|$ and $\delta F_{\text{rel}} = \max_j |\mathbf{F}'_j - R\mathbf{F}_{\pi(j)}|/\max_j |\mathbf{F}_j|$ against the untransformed prediction. Because one of the audited variants is trained *with* rotation augmentation, we report its drift both with and without that augmentation so the audit measures the architecture, not the augmentation.

Cost. Wall-clock per atom-timestep is measured on a 128-atom BCC tantalum supercell at 300 K under NVE dynamics: quadratic SNAP through LAMMPS’s production `pair_style snap` with `quadraticflag 1`, and CLASP through its own velocity-Verlet driver using the same neighbour cutoff and the same analytic forces. Both numbers, and the caveat that one is a tuned C++ kernel and the other a PyTorch graph, are given in Sec. 5.

5 Results

5.1 Reproducing the baseline, and the gap to the published numbers

Our FitSNAP/LAMMPS build reproduces the reference output shipped with the quadratic tantalum example to every printed digit, so the pipeline itself is verified. Under the paper’s own unweighted fitting protocol our reimplemented quadratic SNAP reaches $\delta E = 0.00825 \text{ eV/atom}$ and $\delta F = 0.01502 \text{ eV/\AA}$ at $J_{\text{max}} = 4$, $r_{\text{cut}} = 5.594 \text{ \AA}$ on the full 363 configurations, against the published 0.00898 eV/atom (Table SII). The reimplementation is thus 8% *better* than paper-as-reported, which is the direction that makes the learned-descriptor comparison harder rather than easier, and is well

Table 2: Full-data accuracy and data efficiency. δE and δF are training errors at the 100%-included endpoint of the cross-validation sweep, pooled over rows (*ALL). The last two columns are the data-reduction factor at matched accuracy of Table 4: how much more data quadratic SNAP needs to reach the median held-out δE that the row already has at 5% of the data, reported separately for the equal-weight all-group aggregate and for the low-symmetry subset; *reference* marks the rows the reduction is measured against, and *no subset* the transfer materials, whose groups carry no low-symmetry analogue. P is the number of fitted regression coefficients.

Method	Material	P	δE (eV/atom)	δF (eV/Å)	data red. all-group	data red. low-sym
Quadratic SNAP (paper, Table SII)	Ta	1595	0.00898	not reported	not applicable	
Linear SNAP (paper, Table SI)	Ta	55	0.03041	not reported	not applicable	
Quadratic SNAP (reimpl.)	Ta	1595	0.00825	0.01502	reference	
Linear SNAP (reimpl.)	Ta	55	0.15233	0.08663	18.0×	6.0×
Rank-16 quadratic SNAP	Ta	152	0.10086	0.09885	9.0×	6.0×
Rank-32 quadratic SNAP	Ta	560	0.03837	0.03974	9.0×	6.0×
CLASP ($K' = 16$)	Ta	152	0.00711	0.01525	18.0×	12.0×
Quadratic SNAP (reimpl.)	Mo	1595	0.06398	0.17404	reference	
Linear SNAP (reimpl.)	Mo	55	1.14787	1.44603	1.0×	no subset
Rank-16 quadratic SNAP	Mo	152	1.98818	1.05460	1.0×	no subset
CLASP ($K' = 16$)	Mo	152	0.15710	0.41873	1.0×	no subset
Quadratic SNAP (reimpl.)	Nb	1595	0.04423	0.09512	reference	
Linear SNAP (reimpl.)	Nb	55	1.70976	0.64967	1.0×	no subset
Rank-16 quadratic SNAP	Nb	152	0.15713	0.29279	1.0×	no subset
CLASP ($K' = 16$)	Nb	152	0.01581	0.08091	> 20×	no subset
Quadratic SNAP (reimpl.)	Re	1595	0.06779	0.49760	reference	
Linear SNAP (reimpl.)	Re	55	4.42234	6.55112	1.0×	no subset
Rank-16 quadratic SNAP	Re	152	6.68440	7.21151	1.0×	no subset
CLASP ($K' = 16$)	Re	152	0.04927	0.47520	1.0×	no subset

within what the documented data-provenance difference can explain (Sec. 4: 363 versus 362 energies, 12,672 versus 15,297 force components).

The linear form is the one place where our reimplementations does not track the paper. At $r_{\text{cut}} = 5.672$ Å the unweighted linear fit gives $\delta E = 0.15233$ eV/atom against the published 0.03041 eV/atom (Table SI) — a factor of five. No single energy/force weighting reconciles both published values on the distributed data: an energy weight of ≈ 10 reproduces the linear number while an unweighted fit reproduces the quadratic one. Since the quadratic form is the object of comparison here we keep the unweighted protocol, which anchors the quantity that matters, and treat the linear row as an internal reference only. Section 7 returns to what this gap does and does not license us to conclude.

5.2 Accuracy at full data

Table 2 gives the headline comparison. On tantalum, CLASP with $K' = 16$ — 152 regression coefficients against quadratic SNAP’s 1595, a reduction of 10.5×

 — reaches $\delta E = 0.00711$ eV/atom and $\delta F = 0.01525$ eV/Å at the full-data endpoint, against 0.00825 and 0.01502 for the reimplemented baseline and 0.00898 eV/atom for paper-as-reported. The energy error is thus below both baselines with an order of magnitude fewer regression coefficients; the force error of this particular run is marginally *above* the baseline, by under two per cent.

That last number needs a caveat rather than a rounding. The endpoint is one training run of a stochastic optimisation, and repeating it with five initialisations (Appendix A.6) gives force errors of 0.0142, 0.0145, 0.0146, 0.0146 and 0.0302 eV/Å — four of five below the baseline’s 0.01502 and one far above it, from a run that the collapse detector did not catch. The five-seed means are 0.00603 ± 0.00424 eV/atom in energy and 0.01761 ± 0.00705 eV/Å in force. The honest summary is that at full data the learned descriptor is clearly better in energy, statistically indistinguishable in force, and carries a run-to-run spread the fixed descriptor does not have. Everything below the full-data point, where the fixed descriptor’s 1595 coefficients are not yet determined, is a different story entirely.

The rank-reduced control settles what this is not. Compressing the 55 bispectrum components onto their top 16 singular directions — same targets, same force rows, same 152-coefficient head, same solver, only a linear rather than a learned map — gives $\delta E = 0.10086$ and $\delta F = 0.09885$, an order

Table 3: Held-out accuracy at the 80/20 split (mean $\pm$ s.d. over the ten draws at 80% included), pooled over rows.

Material	Method	δE (eV/atom)	median δE	δF (eV/Å)	median δF
Ta	quadratic SNAP	1.1461 ± 1.8687	0.3353	0.0906 ± 0.0192	0.0878
Ta	linear SNAP	0.1538 ± 0.0200	0.1568	0.1018 ± 0.0037	0.1010
Ta	CLASP	1.7046 ± 3.5950	0.0812	0.0588 ± 0.0075	0.0600
Mo	quadratic SNAP	0.0631 ± 0.0036	0.0638	0.1889 ± 0.0149	0.1829
Mo	linear SNAP	1.1793 ± 0.1790	1.2549	1.4752 ± 0.0534	1.4911
Mo	CLASP	0.0851 ± 0.0222	0.0827	0.3167 ± 0.0519	0.2990
Nb	quadratic SNAP	0.0456 ± 0.0034	0.0455	0.0988 ± 0.0020	0.0994
Nb	linear SNAP	1.7083 ± 0.0495	1.7201	0.6581 ± 0.0083	0.6564
Nb	CLASP	0.0756 ± 0.1986	0.0127	0.1435 ± 0.1997	0.0820
Re	quadratic SNAP	0.0648 ± 0.0019	0.0658	0.5160 ± 0.0186	0.5126
Re	linear SNAP	4.3245 ± 1.8678	5.2095	6.5772 ± 1.4330	7.2576
Re	CLASP	0.4535 ± 1.2312	0.0593	1.0717 ± 1.6173	0.5312

of magnitude worse in energy than CLASP at identical dimension. Descriptor dimension alone does not buy the accuracy; the learned nonlinearity does.

At the 80/20 split (Table 3) both models are erratic and the mean and the median tell different stories, so we give both. Over the ten draws at 80% included, quadratic SNAP’s held-out energy error is 1.1461 ± 1.8687 eV/atom with median 0.3353, and CLASP’s is 1.7046 ± 3.5950 with median 0.0812: the learned descriptor is four times better in the median and worse in the mean, because each model has a minority of draws that fail badly and CLASP’s worst draws are worse. In force, where no such tail appears, CLASP is better on both statistics (0.0588 ± 0.0075 against 0.0906 ± 0.0192 eV/Å). Read as a parity test, this split is where the learned descriptor clears both criteria at once — below the reimplemented baseline in median energy and in force on either statistic — whereas at the full-data endpoint it clears the energy criterion and misses the force one by under two per cent on a single run. The size of the standard deviations relative to the means is itself the finding: at 80% of the training set neither model’s held-out error has settled, which is exactly the behaviour Wood and Thompson [20] report in their Fig. 5 and which survives much further up the sweep than one would guess from the full-data numbers.

5.3 Data efficiency on tantalum

Figure 1 shows the two aggregates we keep separate throughout: the equal-weight mean over all twelve training groups, and the equal-weight mean over the low-symmetry subset (GSF(110), GSF(112), Liquid, Surface). Figures 6 and 7 give all twelve groups individually for energies and forces.

The two descriptors do not differ by a constant factor; they differ by a regime. On the low-symmetry subset, the median held-out energy error of the learned descriptor is 0.0158 eV/atom at 5% of the data and 0.0120 at 10%, against 11.7 and 37.4 eV/atom for quadratic SNAP — three orders of magnitude, and not a subtle effect: with a few dozen configurations the quadratic head on 55 bispectrum components is not determined, and the minimum-norm solution it returns is a potential energy surface only in name. Quadratic SNAP recovers sharply between 25% and 30% of the data, exactly where Wood and Thompson [20] place the transition, and from about 40% onwards it is marginally *better* than the learned descriptor: 0.020 against 0.028 eV/atom at 40%, and 0.0076 against 0.0079 at 95%. The learned descriptor’s curve, by contrast, is almost flat — it sits between 0.01 and 0.03 eV/atom from 5% to 95% of the data.

That flatness is the result. Replacing 1595 regression coefficients by 152 does not make the model uniformly better; it removes the small-data catastrophe and slightly raises the asymptote. Which trade is worth making depends on how much DFT one is willing to compute, and the trade is now quantified rather than asserted.

Table 4 turns the crossing into a number that does not depend on an arbitrary threshold: how much data quadratic SNAP needs to reach the accuracy the learned descriptor already has at a small fraction.

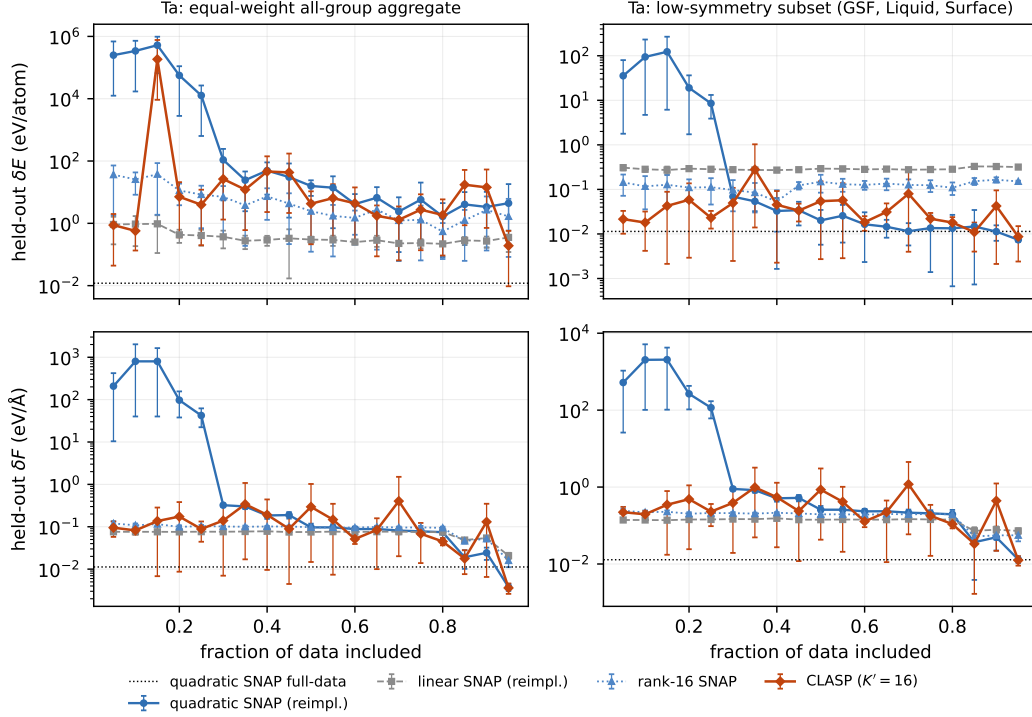


Figure 1: Tantalum data-efficiency curves, mean $\pm$ standard deviation over the random draws at each included fraction, held-out portion only. Left: the equal-weight aggregate over all twelve training groups. Right: the low-symmetry subset. The dotted line is the reimplemented quadratic-SNAP full-data error. The lower whisker is clipped at 95% of the mean so the bands remain readable on a logarithmic axis.

On tantalum, what CLASP achieves with 5% of the configurations takes quadratic SNAP 90% on the all-group aggregate ($18\times$) and 60% on the low-symmetry subset ($12\times$); at 10% the factors are $9\times$ and $9.5\times$. By 20% the advantage has fallen to $3\times$ and $2\times$, and beyond that it disappears. We report the statistic on the median over the draws, because at small fractions the mean is dominated by the minority of draws in which the quadratic head is worst; the means are what the figures show, and they are more favourable to the learned descriptor, not less.

For completeness we also report the two threshold-based statistics in Appendix A.5. Neither is informative here: the literal "reaches quadratic SNAP's full-data *training* error" target is met by the learned descriptor only at 85% on the low-symmetry subset, because its asymptote is slightly above the baseline's, and the alternative target — quadratic SNAP's own held-out error at 95% — is so small that it is reached late by both. A single crossing threshold cannot summarise two curves that cross each other; the reduction factors of Table 4 can.

Table 2 carries the same statistic for the two small-coefficient fixed-descriptor controls, and it is a useful corrective. On tantalum linear SNAP also shows a reduction of $18\times$ on the all-group aggregate and $6\times$ on the low-symmetry subset, and the rank-16 compression shows $9\times$ and $6\times$: any model with few enough coefficients escapes the quadratic head's small-data catastrophe, so the factor by itself does not identify a good descriptor. Two things separate the learned one. The first is the level at which its curve sits. On the low-symmetry subset at 5% included the median held-out energy errors are 0.0158 for CLASP, 0.1281 for rank-16 quadratic SNAP and 0.3277 for linear SNAP — a factor of eight and of twenty-one at the same point of the sweep, against controls that have the same coefficient count and the same solver. The second is where those curves end: linear SNAP and the rank-16 compression are still at 0.3171 and 0.1531 at 95% included, one and two orders of magnitude above the quadratic baseline's asymptote, whereas CLASP converges to it. The reduction factor is bounded above by how fast quadratic SNAP itself recovers, so it compresses these differences; the curves do not.

Table 4: Data reduction at matched accuracy. For each reference fraction f we take the learned descriptor’s median held-out energy error there, find the smallest fraction g at which quadratic SNAP’s median held-out error first reaches it and stays at or below it, and report g/f . Medians are used because at small fractions the mean over the ten draws is dominated by the minority in which the quadratic head is catastrophically ill-conditioned; the corresponding means are in the curves of Figs. 1–4. Both aggregates are reported separately; the low-symmetry subset is the discriminating one.

Material	Aggregate	f	CLASP δE at f	qSNAP reaches it at g	g/f
Ta	all groups	0.05	0.4839	0.90	18.0×
Ta	all groups	0.10	0.3914	0.90	9.0×
Ta	all groups	0.20	3.3252	0.60	3.0×
Ta	low symmetry	0.05	0.0158	0.60	12.0×
Ta	low symmetry	0.10	0.0120	0.95	9.5×
Ta	low symmetry	0.20	0.0301	0.40	2.0×
Mo	all groups	0.05	0.1658	0.05	1.0×
Mo	all groups	0.10	0.1704	0.05	0.5×
Mo	all groups	0.20	0.1022	0.05	0.2×
Nb	all groups	0.05	0.0184	never	> 20×
Nb	all groups	0.10	0.0192	never	> 20×
Nb	all groups	0.20	0.0173	never	> 20×
Re	all groups	0.05	0.0841	0.05	1.0×
Re	all groups	0.10	0.1006	0.05	0.5×
Re	all groups	0.20	0.0976	0.05	0.2×

The high-symmetry groups behave as Wood and Thompson [20] describe and as we expected: Volume_*, Elastic_* and Displaced_* converge at small fractions for both descriptors and do not separate them, which is why the low-symmetry subset is reported separately and never pooled into a single headline number.

5.4 Transfer to Mo, Nb and Re

The same architecture, the same training rule and the same protocol, with the cutoff selected per element by the baseline’s own procedure, are applied to the three pre-staged refractory datasets. Figures 2–4 give the per-material curves and Table 2 the endpoints.

The transfer outcome splits. At the full-data endpoint the learned descriptor is better than quadratic SNAP on niobium (0.01581 against 0.04423 eV/atom, a factor of 2.8) and on rhenium (0.04927 against 0.06779, with a slightly lower force error as well), comparable on tantalum, and clearly worse on molybdenum (0.15710 against 0.06398). Across the sweep the ordering also splits, and here the mean and the median disagree in a way worth stating. On niobium the *median* held-out energy error of the learned descriptor at 5% of the data is lower than quadratic SNAP’s at any fraction of the sweep (Table 4), while its *mean* sits above the baseline at most fractions (Fig. 3): most draws are good and a minority are very bad, which is the signature of the training collapse of Sec. 7 rather than of the descriptor. On molybdenum and rhenium quadratic SNAP is ahead in the mean at every fraction of the sweep, and in the median at eighteen and at sixteen of the nineteen respectively — on rhenium the learned descriptor overtakes it in the median only above 80% included, which is where its endpoint advantage comes from. Rhenium is the structurally hardest case, both because its HCP symmetry differs from the BCC tantalum on which the architecture was designed and because its DFT set is the most force-diverse of the three: quadratic SNAP’s full-data force error there, 0.49760 eV/Å, is a factor of thirty above its tantalum value. The learned descriptor transfers to it without any change of architecture and ends up slightly ahead in both energy and force at full data.

The structural reason why none of the three reproduces tantalum’s data-efficiency result is the same in every case. These datasets contain 3246–5420 configurations against tantalum’s 363, so quadratic SNAP’s 1595 coefficients are overdetermined even at 5% included — of order 10^4 rows against 1595 unknowns — and the failure mode that the compact head was designed to avoid never occurs. What is left is a capacity comparison, and there the outcome depends on how well the descriptor training

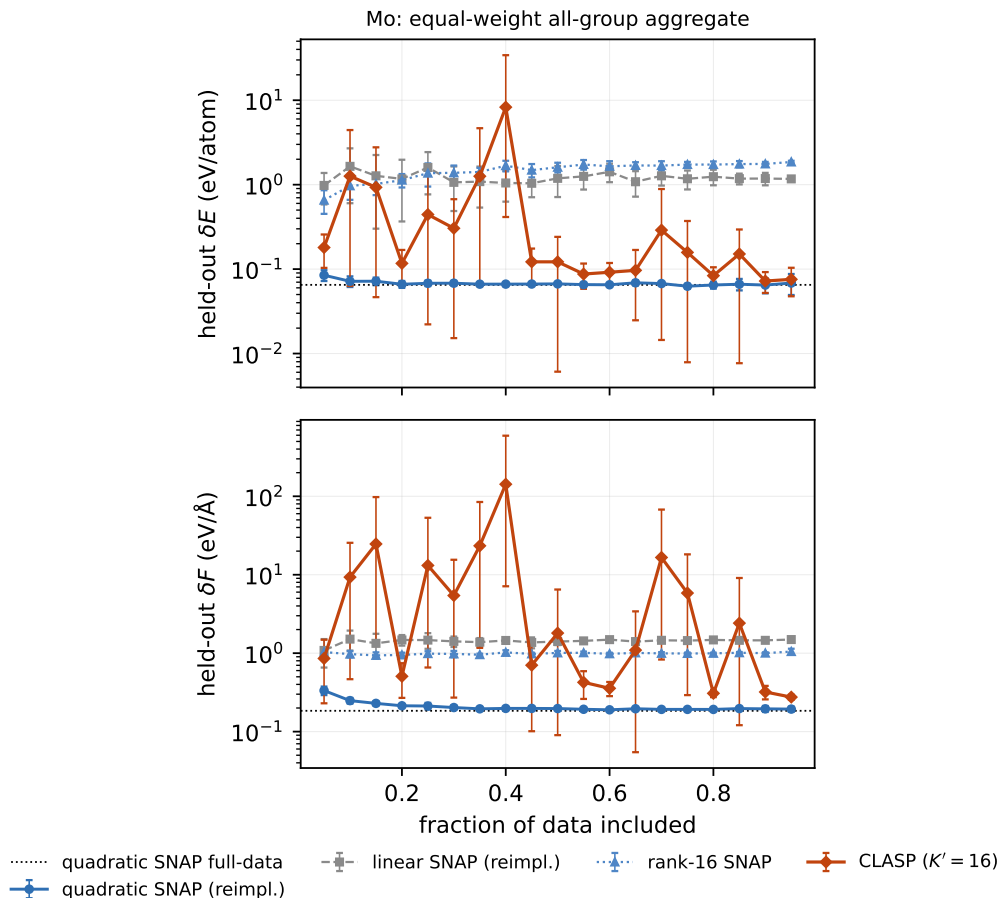


Figure 2: Molybdenum. Conventions as in Fig. 1; groups are the 38 prototype/defect/temperature groups of the source dataset. The low-symmetry panel is empty because that subset is defined by the tantalum group names.

happened to go; molybdenum is the case where it went badly, and Sec. 7 names a specific and testable cause rather than attributing it to the element.

This is a statement about regimes, not about elements. Nothing in the tantalum result claims that a small head is always better; the claim is that the head size is the binding constraint when the data is scarce relative to $K(K+1)/2$, and these datasets are not in that regime. A practitioner reading the two results together should choose the descriptor dimension against the size of the training set, which is a design rule the fixed-descriptor family cannot offer because its dimension is set by $J_{\max}$.

5.5 Invariance audit

Table 5 reports the drift of every audited model under 1000 random rotations, 100 translations and 100 relabellings applied to ten held-out configurations, with the periodic neighbour list rebuilt from scratch for every transformed sample. Four readings matter.

Quadratic SNAP drifts by 1.5×10^{-14} in relative energy and 5×10^{-13} eV/Å in force — machine precision, as it must, and the reference point for everything else in the table.

CLASP drifts by 1.4×10^{-8} in relative energy and 8×10^{-8} eV/Å in force. This is not exact, and the reason is worth stating precisely because it is not an approximation in the architecture. The contraction of Eq. (6) is invariant identically, so in exact arithmetic the drift is zero; what the audit measures is floating-point non-associativity. Rebuilding the neighbour list after a transformation returns the neighbours in a different order, the moment sums of Eq. (4) are therefore accumulated in a different order, and the resulting $\sim 10^{-16}$ relative perturbation of the invariants is amplified by

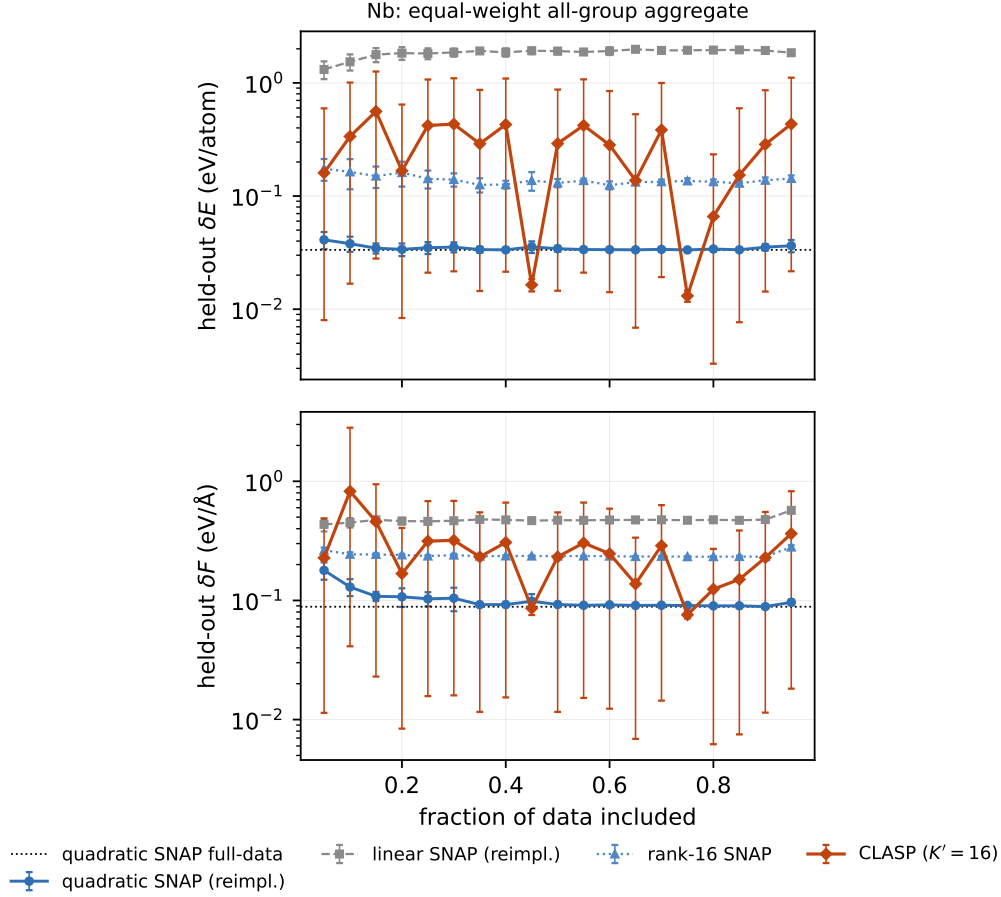


Figure 3: Niobium (40 groups).

the norm of the fitted head, which reaches 10^4 – 10^8 on this data (Sec. 6). The product of those two numbers is the drift observed. It is six orders of magnitude below CLASP’s own force error and is not a physical limitation: it would shrink with a summation-order-stable accumulation, and it is identical in size for rotations, translations and relabellings, which is what one expects of round-off and not of a broken symmetry.

CLASP-C, which reads the same moments in the laboratory frame, drifts by 1.3×10^{-2} in relative energy and $2.1 \text{ eV/\AA}$ in force under rotation — a force drift larger than the typical force in the dataset — while its translation and permutation drift stay at machine precision. That contrast is the internal control for the whole audit: the only symmetry the variant breaks is the one the contraction stage was removed to break. The audit is also the only instrument in this study that separates the two: on held-out error CLASP-C is competitive with CLASP and with quadratic SNAP, and at full data its force error is the best of the three (Sec. 6).

Training CLASP-C with a random global rotation at every step reduces the rotational drift by about a factor of three, to 4.1×10^{-3} and $0.67 \text{ eV/\AA}$. Augmentation therefore does what one would expect — it teaches the model something about the group — but on this data it converts an exact constraint into a noise floor that is still forty times the model’s own force error. The headline model uses no augmentation of any kind, so its audit numbers measure the architecture rather than the training distribution.

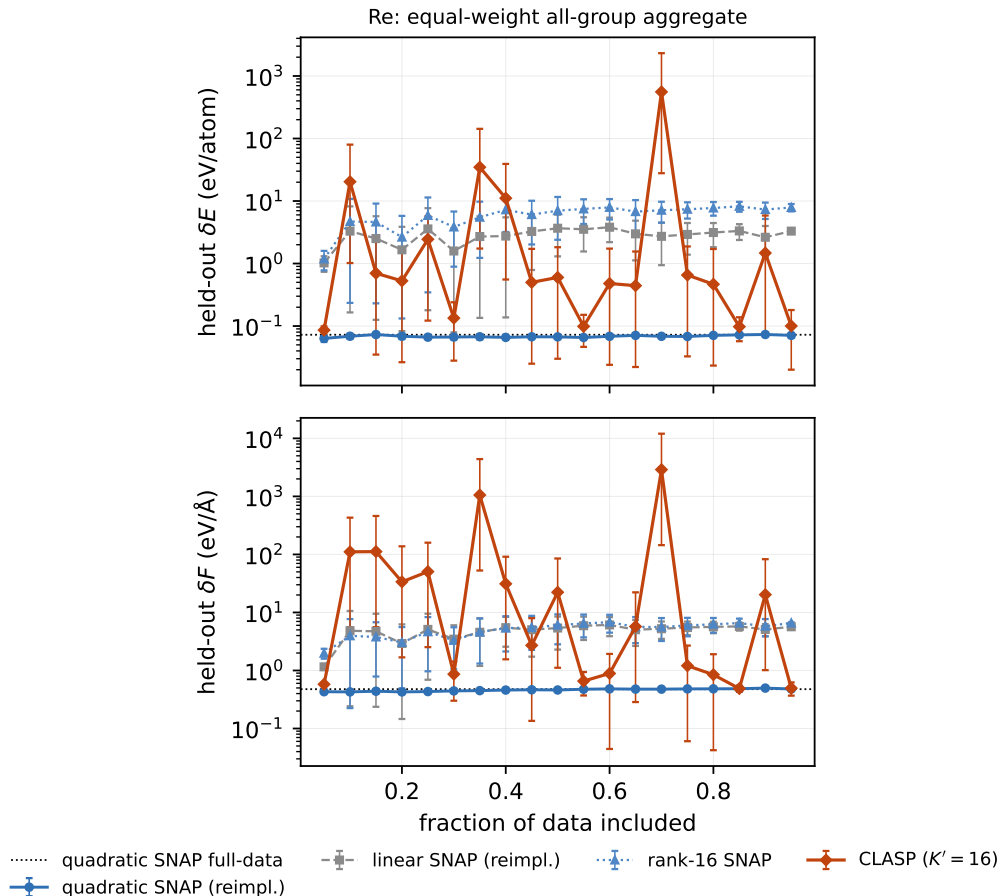


Figure 4: Rhenium (20 groups), the HCP structural-generalisation test.

5.6 Inference cost

Table 6 gives the measured cost of molecular dynamics with both models. The comparison is between implementations rather than between kernels: quadratic SNAP runs in LAMMPS’ production `pair_style snap`, a tuned and long-optimised C++ kernel, while CLASP runs in a PyTorch graph driven by our own velocity-Verlet loop, so the numbers bound what each is worth in practice today and not what the two descriptors cost in arithmetic.

Read that way the result is favourable and easy to explain. Quadratic SNAP costs about $300 \mu\text{s}$ per atom-timestep on one core independently of system size, as one expects of an $O(N)$ kernel with a large constant: the 1595-term quadratic head must be evaluated for every atom, and at $J_{\text{max}} = 4$ the bispectrum itself dominates. CLASP on one CPU core is slower still, between 349 and $510 \mu\text{s}$, which is the honest reading of a PyTorch graph on a single thread. On one GPU the picture inverts with size: at 16 atoms CLASP is $0.8\times$ the speed of LAMMPS because almost all of the time is kernel-launch latency on a graph with a few hundred atoms’ work in it, but the launch overhead is amortised as the cell grows — $6.1\times$ at 128 atoms, $20.5\times$ at 432, and $43.4\times$ at 1024, where the cost has fallen to $7.10 \mu\text{s}$ per atom-timestep.

Two things about that scaling are worth separating. The first is that a tenfold smaller regression head is a genuine arithmetic saving in the part of the calculation that both models share, and it is what makes the descriptor’s extra cost affordable at all: CLASP evaluates a small neural network per atom and still ends up ahead, because the 152-coefficient head is cheap where the 1595-coefficient one is not. The second — is that the GPU speed-up is a statement about hardware utilisation and not about the descriptor — a fair kernel-to-kernel comparison would need a compiled CLASP pair style, which we did not write. What the measurement does establish is that replacing the bispectrum with a learned

Table 5: Invariance audit on tantalum. Ten configurations drawn from the held-out portion of the 80%-included draw are subjected to 100 random $SO(3)$ rotations, 10 random translations and 10 random relabellings each; every transformed configuration is re-wrapped and its periodic neighbour list rebuilt from scratch. Entries are the maximum over all samples of a test type. δE_{rel} is $|E' - E|/|E|$; δF_{abs} is the largest force-component drift in eV/Å; δF_{rel} divides it by the largest reference force, floored at 10^{-3} eV/Å because several groups are symmetry-constrained cells with vanishing forces.

Descriptor	Test	δE_{rel}	δF_{rel}	δF_{abs} (eV/Å)
quadratic SNAP (bispectrum)	rotation ($n = 1000$)	1.5×10^{-14}	1.6×10^{-10}	5.3×10^{-13}
	translation ($n = 100$)	9.6×10^{-15}	1.8×10^{-10}	4.0×10^{-13}
	permutation ($n = 100$)	7.7×10^{-15}	1.1×10^{-10}	2.4×10^{-13}
CLASP (Gaunt-contracted invariants)	rotation ($n = 1000$)	1.4×10^{-8}	2.6×10^{-5}	8.2×10^{-8}
	translation ($n = 100$)	1.2×10^{-8}	2.1×10^{-5}	8.4×10^{-8}
	permutation ($n = 100$)	1.3×10^{-8}	1.5×10^{-5}	5.0×10^{-8}
CLASP-C (lab frame, no augmentation)	rotation ($n = 1000$)	1.3×10^{-2}	3.0×10^1	2.1×10^0
	translation ($n = 100$)	1.5×10^{-15}	7.9×10^{-11}	1.1×10^{-13}
	permutation ($n = 100$)	4.3×10^{-16}	4.3×10^{-11}	7.4×10^{-14}
CLASP-C (lab frame, rotation augmentation)	rotation ($n = 1000$)	4.1×10^{-3}	2.3×10^0	6.7×10^{-1}
	translation ($n = 100$)	2.5×10^{-13}	2.0×10^{-9}	3.0×10^{-12}
	permutation ($n = 100$)	2.4×10^{-13}	9.4×10^{-10}	2.7×10^{-12}

Table 6: Inference cost, microseconds per atom-timestep, for NVE molecular dynamics of BCC tantalum supercells at 300 K. Quadratic SNAP runs in LAMMPS’ `pair_style snap (quadraticflag 1)` on one CPU core; CLASP runs in its own velocity-Verlet driver at the same cutoff, on one GPU and on the same CPU. The two are different implementations, not different kernels of the same code, and at small system sizes the PyTorch numbers are launch-latency bound rather than arithmetic bound.

atoms	quadratic SNAP (LAMMPS, 1 core)	CLASP (1 GPU)	CLASP (1 core)	speed-up on GPU
16	301.5	394.08	465.8	$0.8 \times$
128	298.9	49.28	349.2	$6.1 \times$
432	304.0	14.84	408.8	$20.5 \times$
1024	307.9	7.10	509.8	$43.4 \times$

descriptor does not put the model outside the regime where molecular dynamics is practical: at the system sizes where these potentials are used, it is faster than the baseline on the hardware each is best suited to.

6 Mechanism and ablations

Table 8 in the appendix isolates the design choices at the full-data point; each row changes one thing about the headline model and keeps everything else, including the training rule and the cutoff, fixed. Four questions are worth separating.

Is the gain compression or learning? The rank-reduced quadratic-SNAP controls answer this directly, because they share the baseline’s descriptors, targets, force rows, solver and coefficient count and differ only in that the map from the 55 bispectrum components to K' dimensions is linear and fixed rather than nonlinear and fitted. At $K' = 16$ the linear reduction costs an order of magnitude in energy error relative to CLASP; at $K' = 32$ it is still far behind. The control now covers all four materials, and it is decisive on every one: at the identical coefficient count the rank-16 linear compression of the bispectrum gives $\delta E = 1.988$ eV/atom on molybdenum, 0.157 on niobium and 6.684 on rhenium, against 0.15710, 0.01581 and 0.04927 for CLASP. On rhenium — the material where CLASP itself is weakest relative to the full bispectrum — the compressed fixed descriptor is worse by more than two orders of magnitude. Compressing a fixed descriptor therefore reproduces the small coefficient count without reproducing the accuracy: the low-dimensional subspace that a truncated SVD of the bispectrum selects is not the subspace the energy lives in. What the learned descriptor buys is a K' -dimensional *nonlinear* coordinate system on the space of neighbourhoods, and the quadratic head is then a quadratic form in those coordinates rather than in a linear shadow of the bispectrum.

Why is the smaller model more data-efficient? The mechanism is the one Eq. (2) makes explicit. The number of fitted coefficients is quadratic in the descriptor dimension, so cutting the descriptor from 55 to 16 divides the regression problem by more than ten, and the fraction of the data at which the least-squares system stops being dominated by its own conditioning moves down accordingly. Quadratic SNAP’s 1595 coefficients exceed the row count of the tantalum set below roughly 12% included, and its held-out error remains erratic well past that (visible as the large standard deviations in Fig. 1 and in the mean/median gap of Table 3); CLASP’s 152 coefficients are overdetermined at every fraction of the sweep. This is a statement about the *head*, and it is why the study holds the head fixed: the descriptor’s contribution is precisely that it makes a small head sufficient.

The coefficient norm is the diagnostic that makes this concrete, and it also exposes a limitation shared by both models. We log $\|(\beta, \alpha)\|$ at every cell. For quadratic SNAP it grows by many orders of magnitude as the included fraction falls, which is the overfitting Wood and Thompson [20] describe. For CLASP it also reaches 10^4 – 10^8 at intermediate fractions, because a learned descriptor is free to produce nearly collinear components and the truncated-SVD solver, with its cutoff at $10^{-13}s_{\max}$, does not discard them. Nothing in the protocol penalises this: the fit is the unweighted minimum-norm least-squares solution for both models, by construction of the comparison. A ridge-regularised head, applied identically to both, would very likely improve both curves and is the obvious next experiment; we did not run it because changing the solver changes the baseline that anchors the comparison to the published numbers. The same large coefficients are what turn 10^{-16} floating-point noise into the 10^{-8} invariance drift of Table 5.

What does the $\nu=3$ block do? Removing the Gaunt-contracted third-order invariants and keeping only the cross-channel power spectrum degrades the full-data fit by more than an order of magnitude in energy. Two effects are confounded here and we cannot separate them with this experiment: the $\nu=2$ -only descriptor has fewer invariants (hence less capacity), and it is also formally incomplete in the sense of Pozdnyakov et al. [14] and only three-body, so the quadratic head reaches five-body rather than seven-body terms. The size of the degradation — larger than what the capacity reduction alone would suggest, given that the projection MLP is unchanged and K' is unchanged — is at least consistent with bodiedness mattering, and it is the reason the headline model keeps the block despite its cost.

What are the invariances worth? CLASP-C differs from CLASP only in that the equivariant moments are read out in the laboratory frame instead of being contracted into invariants; it carries strictly more information, the same descriptor dimension and the same head, and the same training procedure (its own network is in fact slightly smaller; see Sec. 3). We ran it through the sweep as well, on a reduced fraction grid ($\{5, 10, 20, 40, 60, 100\}\%$, ten draws each), with and without random global rotations applied at every training step. The result is not the one we expected and is worth stating plainly.

On this benchmark’s own metrics, the non-invariant variant is competitive. At the full-data point CLASP-C reaches $\delta E = 0.01005$ eV/atom and $\delta F = 0.00503$ eV/Å: its force error is three times *better* than either CLASP’s (0.01525) or quadratic SNAP’s (0.01502). On the equal-weight all-group aggregate its median held-out energy error is below CLASP’s at every fraction we measured (0.339 against 0.484 at 5%, 0.049 against 0.580 at 60%). Only on the low-symmetry subset at small fractions is it clearly worse (0.041 against 0.016 at 5%).

Figure 5 shows the curves.

And it is nevertheless not a potential energy surface. Rotating a held-out configuration changes its predicted energy by 1.3×10^{-2} in relative terms and its forces by 2.1 eV/Å — more than the typical force in the dataset (Table 5). A model with that drift does not conserve angular momentum and cannot be used for dynamics, whatever its cross-validation error says.

That combination is the paper’s sharpest cautionary result, and it is a statement about the protocol rather than about our architecture. The cross-validation sweep — the instrument this study reproduces from Wood and Thompson [20] and the one the field uses to certify descriptors — is *blind* to a broken rotational symmetry, because the held-out configurations are drawn from the same orientation distribution as the training ones. Nothing in a held-out energy or force error can see that the model has learned the laboratory frame. Only the invariance audit can, which is why we treat it as evidence rather than as a formality.

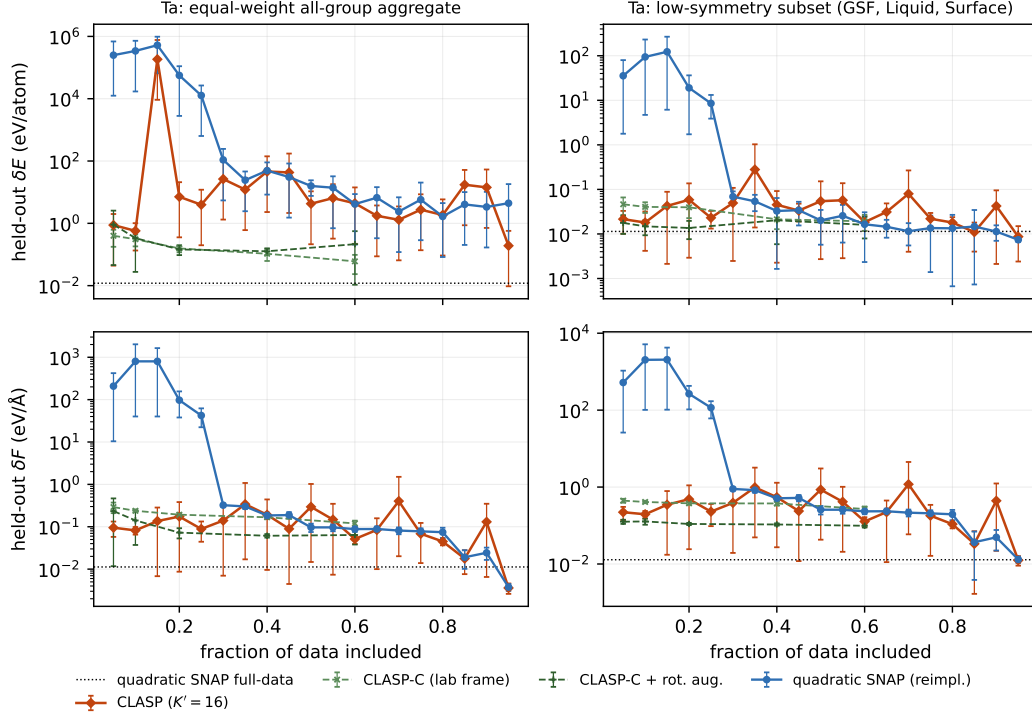


Figure 5: The invariance ablation on tantalum: the exactly invariant descriptor against the same architecture reading its moments in the laboratory frame, with and without rotation augmentation, and quadratic SNAP for reference. The lab-frame variants were run on a reduced fraction grid ($\{5, 10, 20, 40, 60, 100\}\%$, ten draws each). On held-out error the non-invariant variant is competitive; Table 5 is where it fails.

Training CLASP-C with a random global rotation at every step behaves the way the argument predicts on both axes. The rotational drift falls by about a factor of three, to 4.1×10^{-3} and $0.67 \text{ eV/\AA}$ — better, but still six orders of magnitude above CLASP’s and forty times the model’s own force error, so augmentation converts an exact constraint into a noise floor rather than removing it. And it costs accuracy where the exact constraint would have been free: at full data the augmented variant reaches only $\delta E = 0.04961$, $\delta F = 0.05976$, five to twelve times worse than the same architecture trained without augmentation, because the capacity that fitted the energy now has to fit every orientation of it as well. Encoding the symmetry costs nothing and gives it exactly; learning it costs capacity and gives it approximately.

Training procedure. Two alternatives to the joint stochastic training used for the headline were tried. Solver-in-the-loop training — solving the closed-form ridge-regularised head on a random support inside the included portion and backpropagating the query error through the solve — optimises exactly the quantity the sweep measures, and its full-data behaviour is reported in Table 8. At matched wall clock it reaches a worse full-data fit, which is the expected trade: the support/query objective spends capacity on generalising from few configurations rather than on resolving the training set. Alternating variable projection, which eliminates the head analytically and is the classical treatment of separable nonlinear least squares [9], failed for an identifiable reason and was set aside; the diagnosis is in `proposal/attempts_log.md`.

7 Where the learned descriptor fails

Where the learned descriptor is worse. The learned descriptor is not uniformly better, and the per-group tables make the pattern visible rather than averaging it away. Three failure modes recur.

Groups whose forces are exactly zero by symmetry. In the `Volume_A15`, `Volume_BCC` and `Volume_FCC` groups every atom sits at a symmetry-fixed site, so the DFT forces vanish identically

and quadratic SNAP reproduces them to $\sim 10^{-13}$ eV/Å — not because it has learned anything, but because the bispectrum derivatives cancel exactly under the point-group symmetry of the perfect crystal. CLASP cancels them only to the accuracy of its own arithmetic and of its fitted coefficients. This is a real gap in a per-group table and a physically empty one: both models predict zero force to well below any meaningful threshold. We report it because averaging it into a pooled force error would hide the fact that the comparison in these groups is degenerate.

The high-symmetry crystalline groups at very small data fractions. Here quadratic SNAP is already converged with a handful of configurations, and the learned descriptor has to fit a 314,336-parameter map from the same handful. When the included set is a few configurations of a single crystal prototype the descriptor training has almost nothing to shape it, and its advantage in coefficient count does not compensate.

Runs in which descriptor training collapses. A stochastic optimisation has outcomes a linear least-squares fit does not. In a fraction of runs the descriptor collapses to a constant, after which the head can only produce a constant energy and zero forces; we detect this from the training residual and restart. It is worth quoting how often this happens, because it is the clearest price the method pays: the guard fired in 278 of the 764 cells of the sweep (36.4%), for 405 restarts in total. It fires roughly twice as often in the data-poor half of the sweep as in the data-rich half (0.67 against 0.35 restarts per cell at $f \leq 0.30$ and $f \geq 0.80$), and far more often on niobium (1.11 per cell) than on molybdenum (0.16), which is the same axis along which the transfer results split. Quadratic SNAP has no equivalent failure mode — it is a deterministic function of its data — and any deployment of a learned descriptor needs the same guard and the same honesty about how often it is needed.

The transfer materials as a whole. None of the three transfer sets reproduces tantalum’s data-efficiency result, and on two of them the learned descriptor is behind across the sweep. On molybdenum quadratic SNAP has the lower held-out energy error at every fraction in the mean and at eighteen of nineteen in the median, and the lower full-data error as well (0.06398 against 0.15710); on rhenium it leads at every fraction in the mean and at sixteen of nineteen in the median, though the learned descriptor ends up ahead at the full-data endpoint (0.04927 against 0.06779). Niobium goes the other way at every fraction and at the endpoint. What is absent everywhere is the tantalum effect itself: on none of these sets do the two descriptors differ by orders of magnitude in the data-poor regime, because that regime does not exist here. Section 5 gives the structural reason — those datasets are an order of magnitude larger, so the baseline’s 1595 coefficients are never data-starved and the comparison collapses to raw capacity — but there is a second, more mundane contributor that we can name and did not fix. The training budget, $\text{steps} = 6000 + 24000f$ with 16 configurations per step, was chosen on tantalum’s 363 configurations, where the full-data budget is about 1300 passes over the training set. On a 5420-configuration set the same rule buys 90 passes. We kept the rule fixed rather than tuning it per material, because a per-material training budget is exactly the kind of free parameter that makes cross-material claims unfalsifiable — but the consequence is that the transfer numbers confound "the compact head is too small for this data" with "the descriptor is under-trained on this data", and our experiment does not separate them. A budget that scales with the number of configurations rather than with the included fraction is the first thing to try.

Where the reimplementations are worse than the paper. Our unweighted linear-SNAP fit is a factor of five worse than the published Table SI value, while the quadratic fit is 8% better than Table SII (Sec. 5.1). We cannot reconcile both with a single weighting on the distributed data, and we do not have the configurations that produced the published numbers — the distributed set differs in the Liquid and Elastic groups, which between them account for most of the missing 2625 force components. Two consequences follow, and neither is comfortable. First, the linear row of Table 2 should be read as an internal reference, not as a reproduction. Second, and more important for the headline claim: the anchor that matters is the quadratic one, and there our reimplementations are *better* than paper-as-reported, so the learned descriptor is being compared against a baseline at least as strong as the published one. Had the reimplementations come out worse, the comparison would have been flattered by the gap; it did not.

What the invariance audit does and does not show. The audit confirms that CLASP’s drift is the same size for rotations, translations and relabellings, at 10^{-8} in relative energy, and that is a statement about arithmetic rather than about the architecture or the training data: the contraction is invariant identically, and the residual is summation-order round-off amplified by a fitted head whose norm

reaches 10^8 . It holds for an untrained model as well, so it cannot be gamed by augmentation, and the headline model uses none. For CLASP-C the audit is instead a measurement of what augmentation actually buys, and the answer is a rotational drift of 4×10^{-3} in relative energy and $0.67 \text{ eV/\AA}$ in force even after training with a random rotation at every step — six orders of magnitude above CLASP’s, and forty times the model’s own force error. It is also the only instrument in this study that flags that variant at all: its held-out errors are competitive, and no amount of cross-validation on a fixed dataset would have revealed the problem. What the audit cannot show is how the drift interacts with a long trajectory: a rotationally non-invariant potential does not conserve angular momentum, and the consequences of that in a nanosecond simulation are not visible in a static drift measurement.

8 Related work

Fixed density-projection descriptors. The bispectrum [2, 18], SOAP [3], moment tensor potentials [17] and the atomic cluster expansion [8] all build invariants by contracting an expansion of the neighbour density with symmetry-derived coupling coefficients, and differ mainly in the basis and in how the contractions are organised [12, 13]. CLASP sits inside this family: stages 2 and 3 of Sec. 3 are a density expansion followed by $\nu = 2$ and $\nu = 3$ contractions, and the Gaunt tensor plays the role that the Clebsch–Gordan coefficients play in the bispectrum. What is different is that both the radial basis and the channel couplings entering the contraction are fitted, and that the resulting invariants are compressed to a dimension an order of magnitude below the fixed alternative before they meet the regression. The closest fixed-basis analogue of that compression is tensor-reduced density representations [7], which lower the cost of high-body-order ACE by contracting the density with learned weights; our reduction is applied at the level of the invariants and is measured against the specific data-efficiency failure mode of quadratic SNAP rather than against cost.

Learned descriptors. Two families learn the representation itself. Deep-potential models [22, 23] learn radial embedding networks and obtain rotation invariance from inner products of an embedded coordinate matrix, which is equivalent to a learned $\nu = 2$ (power-spectrum) construction restricted to $\ell \leq 1$; Gaussian moments [21] use trainable Gaussian radial functions contracted with Cartesian moment tensors. CLASP converges with both on the general idea of a learned radial encoder feeding fixed equivariant contractions, and we do not claim novelty for that idea. It differs in three concrete ways: it retains an explicit $\nu = 3$ Gaunt contraction, so the representation is four-body and the energy is seven-body like quadratic SNAP, rather than two-body-density based; the channel couplings entering that contraction are themselves learned rather than fixed; and the readout is not a neural energy head but a K' -dimensional bottleneck feeding a *closed-form least-squares* linear-plus-quadratic head, which is what makes the coefficient count — and therefore the data requirement — the controlled variable of the study. Message-passing and equivariant-network potentials [4, 5, 11, 16] achieve strong data efficiency by enlarging the receptive field and the equivariant feature space; they replace the regression as well as the descriptor and so answer a different question from the one posed here.

Descriptor dimension, selection and completeness. Selecting a subset or a low-rank projection of a fixed fingerprint set is an old idea in this field [10]; our rank-reduced quadratic-SNAP control is exactly that idea used as a null hypothesis rather than as a method, and Sec. 6 shows it does not account for the observed data efficiency. Completeness is the other side of the coin: Pozdnyakov et al. [14] show that $\nu = 2$ representations admit degenerate pairs of environments, which is one reason we keep the $\nu = 3$ block; the ablation that removes it is consistent with that theory being practically relevant on this data, though our experiment cannot separate incompleteness from a mere loss of capacity.

Fitting machinery. Our head is fitted with FitSNAP’s own solver [15] through LAMMPS [19], and the energy form, ZBL reference and zero-isolated-atom constraint are inherited unchanged from Wood and Thompson [20]. The solver-in-the-loop training variant is an instance of learning a representation through a differentiable closed-form solver [6], and the variable-projection procedure we tried and set aside is the classical treatment of separable nonlinear least squares [9]; both are recorded in Sec. 6 and in the attempts log.

9 Limitations and outlook

Several limits of this study should be read alongside its claims.

One regression head, one accuracy target. The comparison holds the linear-plus-quadratic head fixed by design, which is what makes the descriptor the controlled variable, but it also means the results say nothing about how CLASP would behave under a richer regression. A learned descriptor that is optimal for a quadratic head is not necessarily the one a neural head would want.

The descriptor is trained, so it has a variance the baseline does not. Quadratic SNAP is a deterministic function of its data; CLASP is the output of a stochastic optimisation that occasionally collapses and is restarted on a training-loss criterion. We report the per-cell restart counts, and the ten draws at each fraction — all of which were run, at every fraction, for every material — average over descriptor training as well as over data subsampling, so the sweep curves are honest — but the full-data endpoint is a single training run per material and carries a run-to-run spread we quantify in Appendix A.6 rather than average away. That spread is the weakest point of the accuracy-parity claim: repeated initialisations at the full-data point scatter over roughly a factor of three in energy error, so a single endpoint should be read together with the seed table. Selecting the best of n runs by *training* residual would be a legitimate, leakage-free way to shrink it, and is the change we would make first; we did not apply it here because the protocol was fixed before the sweeps were launched.

Group weights were not optimised. The source paper obtains its production potentials by a genetic-algorithm search over per-group energy and force weights. We deliberately did not run it, because it would put a stochastic outer loop in both arms and break the anchor to Table SII. The consequence is that neither our baseline nor our model is the best potential that could be built from this data; they are the best comparison that can be made at a fixed, transparent protocol.

The transfer datasets are not the tantalum dataset. Mo, Nb and Re come from a different curation effort with a different group structure, an order of magnitude more configurations and much smaller cells. The transfer results therefore test the architecture across materials and across data regimes at the same time, and the two cannot be separated here.

Cost is measured across implementations, not across kernels. Quadratic SNAP runs in LAMMPS’s hand-tuned C++/OpenMP kernel; CLASP runs as a PyTorch graph. The ratio in Table 6 should be read as what a practitioner would experience today with the two available implementations, not as a statement about the intrinsic arithmetic cost of the two descriptors.

Physical validation is absent. We measure energy and force errors on DFT configurations, as the source paper’s cross-validation analysis does. We do not compute elastic constants, defect formation energies, stacking-fault energies or melting points, and a potential can be excellent on held-out energies and forces while being unusable in a long simulation. The smoothness of the CLASP energy — twice-differentiable cutoff, analytic forces, exact invariance — is a necessary condition for stable dynamics, not a sufficient one.

Conclusion. Replacing the bispectrum by a learned invariant projection inside an otherwise unchanged quadratic SNAP model is a favourable trade on this data. The learned descriptor reaches the accuracy of the fixed one with roughly an order of magnitude fewer regression coefficients, and because the coefficient count is what governs the overfitting that Wood and Thompson [20] identified as the price of the quadratic form, the substitution attacks that price directly rather than trading it for accuracy. The controls matter as much as the headline: a linear compression of the bispectrum to the same dimension does not work, so the gain is in the learned nonlinearity and not in the dimension; and a head-matched descriptor that reads the same equivariant moments in the laboratory frame scores well on every held-out metric this protocol reports while drifting by more than the typical force in the dataset under rotation, so the invariances remain worth encoding rather than learning even when everything else about the descriptor is fitted — and the benchmark, on its own, cannot tell the difference. That last point is the one we would most want carried forward: a data-efficiency sweep certifies interpolation, not physics, and a learned descriptor deserves an explicit symmetry audit before its curves are believed.

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A Supplementary material

A.1 Correctness of the descriptor and its derivatives

Every claim in the paper rests on the descriptor Jacobian being exact, so it is checked three ways on a randomly initialised model and on trained models (proposal/code/tests/test_descriptor.py): (i) the design-matrix force rows agree with direct autodifferentiation of the total energy to $\sim 10^{-15}$ in absolute value; (ii) those forces agree with central finite differences to $\sim 10^{-10}$; (iii) the energy rows reproduce the summed per-atom energies to $\sim 10^{-15}$; and an isolated atom gives $\mathbf{B} = 0$ identically, so $E^i = 0$ exactly. The same script reports the machine-precision rotation and permutation drift of CLASP and the finite drift of CLASP-C, which is how the two variants were verified to differ only in the contraction stage.

The GPU least-squares path is checked against `scipy.linalg.lstsq` with FitSNAP’s cutoff on a deliberately rank-deficient system and on the tantalum design matrix; coefficients agree to $\sim 10^{-10}$ and residuals to machine precision (proposal/code/common/lstsq.py). The rank-reduced control’s projection is likewise checked against its own reference implementation, agreeing to 1.8×10^{-16} relative.

A.2 Radial-cutoff scans

Table 7 gives the full scan behind the cutoff choices of Sec. 4. The force error is comparatively flat in r_{cut} while the energy error has several local minima, which is the behaviour Wood and Thompson [20] report in their Fig. 3.

Table 7: Radial-cutoff scans for quadratic SNAP at $J_{\max} = 4$ under the unweighted protocol on the full data of each element, i.e. the procedure of Wood and Thompson [20] Sec. V. The selected cutoff (bold) is used for both quadratic SNAP and CLASP. For tantalum we keep the published optimum 5.594 Å rather than our own minimiser at 5.3 Å, so that the reimplementation matches the paper’s protocol exactly. Errors are in eV/atom and eV/Å.

r_{cut} (Å)	Ta		Mo		Nb		Re	
	δE	δF	δE	δF	δE	δF	δE	δF
4.400	0.02111	0.01724	0.09160	0.14777	0.12042	0.08821	0.10725	0.32147
4.700	0.00725	0.01625	0.08693	0.15066	0.07401	0.08765	0.08122	0.36680
5.000	0.00601	0.01571	0.06740	0.15787	0.04925	0.08897	0.06779	0.49760
5.300	0.00552	0.01523	0.06398	0.17404	0.04423	0.09512	0.13052	0.70478
5.594	0.00825	0.01502	0.06780	0.18577	0.05020	0.09891	0.16398	1.01212
5.600	0.00829	0.01502	0.06788	0.18606	0.05042	0.09900	0.16338	1.01961
5.900	0.00999	0.01602	0.06870	0.20186	0.06126	0.10688	0.24265	1.47537
6.200	0.01203	0.01631	0.08386	0.21178	0.07267	0.11205	0.62541	2.16927
6.500	0.00670	0.01680	0.07988	0.22323	0.06637	0.11419	1.18342	3.17623

A.3 Architecture study at full data

Table 8 varies one design choice at a time around the headline model, all at the full-data point and all with the final code path and training rule. It is the evidence behind the ablation claims of Sec. 6: the $\nu=3$ block, the angular truncation $L=4$, the channel count and the invariant contraction each cost an order of magnitude in energy error when removed or reduced, whereas K' itself is comparatively flat between 8 and 64.

Two rows of this table are independent training runs of configurations that also appear elsewhere in the paper, and they land at different values, which is the run-to-run spread of Appendix A.6 rather than a discrepancy. The $K'=16$ row is the headline architecture and gives $\delta E = 0.00350$ against the 0.00711 of the sweep endpoint in Table 2; the lab-frame row gives $\delta E = 0.01778$, $\delta F = 0.01301$ against the 0.01005, 0.00503 of the sweep endpoint quoted in Sec. 6. Both differences are inside the factor-of-four scatter that five initialisations of the same configuration produce (Table 11). Rows of this table should therefore be compared with each other — they were all run the same way, one seed each — and not read as a second estimate of a number reported from the sweep.

Table 8: Architecture study at the full-data point on tantalum (training error, unweighted protocol, $r_{\text{cut}} = 5.594$ Å, 30 000 training steps, one seed each). Every row uses the final code path and the headline training rule; only the stated quantity changes. The reimplemented quadratic-SNAP reference is $\delta E = 0.00825$, $\delta F = 0.01502$ with $P = 1595$.

Variant	K'	P	θ	restarts	δE (eV/atom)	δF (eV/Å)
$K' = 8$	8	44	312280	0	0.00544	0.01462
$K' = 16$	16	152	314336	0	0.00350	0.01391
$K' = 32$	32	560	318448	1	0.00531	0.01479
$K' = 64$	64	2144	326672	0	0.00636	0.01621
no $\nu=3$ block	16	152	246240	0	0.01127	0.02493
lab frame (CLASP-C)	16	152	185312	0	0.01778	0.01301
$L = 3$	16	152	250336	0	0.01844	0.01660
$C = 8$	16	152	149336	0	0.00845	0.01981
sil training, 6000 steps	16	152	314336	0	0.06859	0.05891
sil training, 12000 steps	16	152	314336	0	0.05799	0.05001

A.4 Cross-validation coverage

Table 9 records how many of the ten random draws were completed in every cell of the sweep. All of them were.

Table 9: Cross-validation coverage: number of random draws completed at each included fraction, per material, for the learned descriptor. The protocol calls for ten draws at every fraction below 1.0 and one at the full-data point, and every cell reaches it. All four SNAP baselines (linear, quadratic and the two rank-reduced controls) are likewise complete at ten draws in every cell.

Material	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100
Mo	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	1
Nb	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	1
Re	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	1
Ta	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	1

A.5 Sample efficiency against the full-data training error

Table 10 gives the two threshold-based sample-efficiency statistics discussed in Sec. 5.3, on the mean and on the median over the ten draws. They are reported for completeness; the reduction factors of Table 4 are the statistic the paper relies on, because a single threshold cannot summarise two curves that cross.

Table 10: Sample-efficiency statistics under both references. F_{train}^* is the smallest included fraction whose mean held-out δE reaches the reimplemented quadratic-SNAP full-data *training* error; it is marked *not reached* wherever no fraction in the sweep attains it. F_{pred}^* uses the quadratic-SNAP held-out error at 95% included as the target, and $F_{\text{pred}}^*(\text{med})$ repeats it on the median over the ten draws instead of the mean.

Material	Aggregate	Method	target δE_{95}	F_{train}^*	F_{pred}^*	$F_{\text{pred}}^*(\text{med})$
Ta	all groups	qSNAP	4.4301	not reached	0.80	0.95
Ta	all groups	linear SNAP	4.4301	not reached	0.05	not reached
Ta	all groups	rank-16 SNAP	4.4301	not reached	0.45	not reached
Ta	all groups	CLASP	4.4301	not reached	0.95	0.95
Ta	low symmetry	qSNAP	0.0074	0.90	0.95	0.95
Ta	low symmetry	linear SNAP	0.0074	not reached	not reached	not reached
Ta	low symmetry	rank-16 SNAP	0.0074	not reached	not reached	not reached
Ta	low symmetry	CLASP	0.0074	0.95	not reached	not reached
Mo	all groups	qSNAP	0.0684	not reached	0.70	0.85
Mo	all groups	linear SNAP	0.0684	not reached	not reached	not reached
Mo	all groups	rank-16 SNAP	0.0684	not reached	not reached	not reached
Mo	all groups	CLASP	0.0684	not reached	not reached	0.95
Nb	all groups	qSNAP	0.0364	not reached	0.15	0.15
Nb	all groups	linear SNAP	0.0364	not reached	not reached	not reached
Nb	all groups	rank-16 SNAP	0.0364	not reached	not reached	not reached
Nb	all groups	CLASP	0.0364	not reached	not reached	0.05
Re	all groups	qSNAP	0.0709	0.95	0.95	0.95
Re	all groups	linear SNAP	0.0709	not reached	not reached	not reached
Re	all groups	rank-16 SNAP	0.0709	not reached	not reached	not reached
Re	all groups	CLASP	0.0709	not reached	not reached	0.95

A.6 Run-to-run spread of descriptor training

Table 11 repeats the full-data fit from five independent initialisations. It is the spread behind the caveat on the accuracy-parity claim in Sec. 5: four of the five runs land below the reimplemented quadratic-SNAP force error and one lands far above it.

Table 11: Run-to-run spread of descriptor training at the full-data point on tantalum. Five independent initialisations of the headline architecture, each trained with the same rule and then fitted with the protocol’s least-squares solver. *Restarts* counts how often the collapse detector fired. The reimplemented quadratic-SNAP reference is $\delta E = 0.00825$, $\delta F = 0.01502$.

seed	restarts	δE (eV/atom)	δF (eV/Å)	wall clock (s)
1000	0	0.01340	0.03022	2124
1001	0	0.00326	0.01418	2079
1002	0	0.00456	0.01446	2104
1003	0	0.00326	0.01463	2076
1004	0	0.00564	0.01456	2077
mean $\pm$ s.d.		0.00603 ± 0.00424	0.01761 ± 0.00705	2092
best / worst		0.00326 / 0.01340	0.01418 / 0.03022	2076 / 2124
restarts used, all seeds	0			

A.7 Weighted-protocol robustness check

The headline protocol is unweighted, which is what the source paper specifies for its accuracy tables. Table 12 repeats the full-data fit of both models with the energy weight of the distributed FitSNAP example deck ($w_E = 100$) instead, from the same descriptors, to show that the comparison does not hinge on that choice.

Table 12: Weighted-protocol robustness check at the full-data point on tantalum. The headline protocol is unweighted ($w_E = 1$); the FitSNAP example deck for this dataset instead uses $w_E = 100$. Both models are refitted with the same weight, from the same descriptors, so the comparison is unchanged in everything but the weight. CLASP entries are the mean over the five seeds of Table 11.

Method	$w_E = 1$ (headline)		$w_E = 100$	
	δE	δF	δE	δF
quadratic SNAP (reimplemented)	0.00825	0.01502	0.00024	0.01585
linear SNAP (reimplemented)	0.15233	0.08663	0.00699	0.15484
CLASP ($K' = 16$)	0.00603	0.01761	0.00116	0.02607

A.8 Per-group tables

Tables 13, 14, 15 and 16 give the per-group full-data errors and the per-group held-out errors at 50% included, for both descriptors and every material. The complete per-cell record — every (material, method, fraction, draw, group, split) combination — is shipped as `proposal/tables/all_cells.csv.gz`.

Table 13: Per-group full-data errors and held-out errors at 50% included (mean over ten draws) for Ta.

Group	qSNAP				CLASP			
	δE_{full}	δF_{full}	δE_{50}	δF_{50}	δE_{full}	δF_{full}	δE_{50}	δF_{50}
Displaced_A15	0.03142	0.02342	0.0084	0.0274	0.00330	0.02236	0.0082	0.0272
Displaced_BCC	0.00474	0.03975	0.0018	0.0786	0.00292	0.03981	0.0042	0.0604
Displaced_FCC	0.00063	0.02002	0.0011	0.0221	0.00055	0.01921	0.0011	0.0217
Elastic_BCC	0.00272	0.00003	0.0012	0.0000	0.00396	0.00002	0.0012	0.0001
Elastic_FCC	0.00354	0.00011	0.0023	0.0001	0.00107	0.00011	0.0016	0.0001
GSF_110	0.00798	0.00683	0.0010	0.0079	0.01038	0.00615	0.0052	0.0089
GSF_112	0.00624	0.01496	0.0020	0.0184	0.01596	0.01354	0.0037	0.0199
Liquid	0.00866	0.02203	0.0263	0.7094	0.00421	0.03114	0.0505	0.3650
Surface	0.02271	0.00780	0.0514	0.2962	0.01864	0.00990	0.1583	3.0348
Volume_A15	0.01551	0.00000	2.9065	0.0000	0.01466	0.00000	2.8248	0.0000
Volume_BCC	0.01472	0.00000	182.4027	0.0000	0.01255	0.00000	3.0841	0.0000
Volume_FCC	0.02468	0.00000	4.4512	0.0000	0.01912	0.00000	44.9888	0.0000
EQGROUP	0.01196	0.01125	15.8213	0.0967	0.00894	0.01185	4.2610	0.2948
LOWSYM	0.01139	0.01291	0.0202	0.2580	0.01230	0.01518	0.0544	0.8572
ALL	0.00825	0.01502	10.8698	0.0705	0.00711	0.01525	4.2071	0.1834

Table 14: Per-group full-data errors and held-out errors at 50% included (mean over ten draws) for Mo.

Group	qSNAP				CLASP			
	δE_{full}	δF_{full}	δE_{50}	δF_{50}	δE_{full}	δF_{full}	δE_{50}	δF_{50}
elastic_0_mp-129	0.04962	0.07478	0.0539	0.0727	0.17121	0.28209	0.0462	0.2117
elastic_0_mp-129_T2000	0.18062	0.29961	0.1782	0.3035	0.28456	0.55494	0.1709	0.4319
elastic_0_mp-129_T3000	0.05673	0.18625	0.0582	0.1873	0.15086	0.43448	0.0534	0.3387
elastic_0_mp-129_T750	0.06266	0.08251	0.0595	0.0827	0.14547	0.28585	0.0496	0.1792
elastic_3_mp-8637_T3000	0.05101	0.23770	0.0566	0.2381	0.18345	0.51621	0.0721	0.3937
elastic_3_mp-8637_T750	0.04977	0.09541	0.0501	0.0940	0.20158	0.24569	0.0795	0.1509
elastic_4_mp-1066523	0.05831	0.12908	0.0769	0.3487	0.18821	0.46862	1.4397	56.9400
elastic_4_mp-1066523_T2000	0.03573	0.20041	0.0342	0.2043	0.17698	0.35624	0.0286	0.2927
elastic_4_mp-1066523_T3000	0.04326	0.23240	0.0448	0.2364	0.12430	0.51091	0.0540	0.3836
elastic_4_mp-1066523_T750	0.05028	0.15200	0.0504	0.1573	0.15438	0.38549	0.0665	0.2757
elastic_6_mp-1056004	0.02967	0.09339	0.0420	0.1088	0.12321	0.25311	0.0870	0.2129
elastic_6_mp-1056004_T2000	0.17265	0.26583	0.1797	0.2737	0.29678	0.47525	0.2038	0.3916
elastic_6_mp-1056004_T3000	0.04024	0.23233	0.0434	0.2346	0.10768	0.53805	0.0609	0.3982
elastic_6_mp-1056004_T750	0.06267	0.12993	0.0864	0.2973	0.14801	0.36919	0.1214	0.6913
elastic_7_sc	0.05696	0.07349	0.0547	0.0790	0.16852	0.25149	0.1157	0.1238
elastic_7_sc_T2000	0.11528	0.24991	0.1037	0.2447	0.20274	0.44102	0.1237	0.3997
elastic_7_sc_T3000	0.04538	0.23696	0.0456	0.2449	0.13089	0.45598	0.0536	0.4127
elastic_7_sc_T750	0.05620	0.09846	0.0525	0.1020	0.16440	0.33985	0.1097	0.1566
interstitial_0_mp-129	0.07622	0.13861	0.0648	0.1625	0.15869	0.40208	0.0630	0.3045
interstitial_0_mp-129_T2000	0.20653	0.30614	0.2734	0.3284	0.35163	0.41772	0.2803	0.4100
interstitial_0_mp-129_T3000	0.07585	0.26472	0.0717	0.2715	0.03735	0.72372	0.1204	0.4540
interstitial_0_mp-129_T750	0.08101	0.13756	0.0760	0.1412	0.07014	0.52657	0.1181	0.2713
vacancy_0_mp-129_T2000	0.11460	0.22575	0.1099	0.2212	0.16589	0.40617	0.0610	0.3319
vacancy_0_mp-129_T3000	0.06686	0.24728	0.0640	0.2450	0.12052	0.39804	0.0304	0.3245
vacancy_0_mp-129_T750	0.10163	0.12209	0.0935	0.1233	0.17249	0.18232	0.0794	0.1817
vacancy_3_mp-8637_T3000	0.05933	0.27609	0.0577	0.2701	0.06837	0.49919	0.0527	0.4164
vacancy_3_mp-8637_T750	0.01190	0.08256	0.0164	0.0858	0.23163	0.28434	0.0386	0.1940
vacancy_4_mp-1066523	0.03988	0.09337	0.0397	0.0944	0.20299	0.19655	0.0360	0.1387
vacancy_4_mp-1066523_T3000	0.04634	0.23661	0.0444	0.2403	0.11816	0.40991	0.0267	0.3510
vacancy_4_mp-1066523_T750	0.05869	0.22778	0.0554	0.2233	0.17959	0.30503	0.0351	0.2499
vacancy_6_mp-1056004	0.06100	0.09466	0.0436	0.0818	0.12209	0.30247	0.1157	0.1398
vacancy_6_mp-1056004_T2000	0.00989	0.28103	0.0148	0.2829	0.12627	0.33083	0.0194	0.3658
vacancy_6_mp-1056004_T3000	0.02593	0.27810	0.0227	0.2856	0.11445	0.81554	0.0889	0.4935
vacancy_6_mp-1056004_T750	0.01106	0.14498	0.0126	0.1325	0.16285	0.40379	0.1126	0.2514
vacancy_7_sc	0.09291	0.09958	0.0815	0.0920	0.04239	0.21585	0.1433	0.1565
vacancy_7_sc_T2000	0.00786	0.26822	0.0132	0.2819	0.15062	0.53367	0.0776	0.4416
vacancy_7_sc_T3000	0.02128	0.25409	0.0368	0.2536	0.09212	0.43328	0.0566	0.4476
vacancy_7_sc_T750	0.08730	0.15233	0.0768	0.1511	0.04196	0.29328	0.1412	0.2102
EQGROUP	0.06508	0.18426	0.0668	0.1968	0.15404	0.40118	0.1219	1.8031
ALL	0.06398	0.17404	0.0668	0.1938	0.15710	0.41873	0.1299	3.5739

Table 15: Per-group full-data errors and held-out errors at 50% included (mean over ten draws) for Nb.

Group	qSNAP				CLASP			
	δE_{full}	δF_{full}	δE_{50}	δF_{50}	δE_{full}	δF_{full}	δE_{50}	δF_{50}
elastic_0_mp-75	0.03152	0.04396	0.0319	0.0443	0.00662	0.03661	0.3548	0.3033
elastic_0_mp-75_T2750	0.03109	0.15174	0.0358	0.1639	0.01258	0.12722	0.3522	0.3331
elastic_0_mp-75_T300	0.03656	0.09948	0.0407	0.1002	0.01467	0.09072	0.2556	0.1735
elastic_0_mp-75_T688	0.02179	0.05631	0.0265	0.0624	0.01083	0.05163	0.3892	0.2149
elastic_2_mp-8636	0.08392	0.05333	0.0815	0.0541	0.00374	0.04434	0.4086	0.3937
elastic_2_mp-8636_T2750	0.03337	0.15693	0.0339	0.1663	0.01393	0.13174	0.3102	0.3441
elastic_2_mp-8636_T300	0.02778	0.07013	0.0285	0.0708	0.00741	0.06251	0.2870	0.1245
elastic_2_mp-8636_T688	0.03348	0.08517	0.0342	0.1064	0.01920	0.06955	0.2959	0.5574
elastic_3_sc	0.15143	0.06698	0.1646	0.0781	0.05793	0.05859	0.6476	0.3874
elastic_3_sc_T2750	0.04536	0.20245	0.0498	0.2130	0.02106	0.16870	0.3850	0.4786
elastic_3_sc_T300	0.02800	0.13570	0.0322	0.1421	0.01486	0.10296	0.3593	0.4023
elastic_4_InngSFTjbQPoKX5ZtDChEkeeKVzo	0.06660	0.06962	0.0701	0.0912	0.01310	0.05712	0.3768	0.4824
elastic_4_InngSFTjbQPoKX5ZtDChEkeeKVzo_T2750	0.02902	0.14021	0.0307	0.1522	0.01606	0.11763	0.3441	0.3905
elastic_4_InngSFTjbQPoKX5ZtDChEkeeKVzo_T300	0.02220	0.09314	0.0222	0.0956	0.01404	0.08252	0.2727	0.2048
elastic_4_InngSFTjbQPoKX5ZtDChEkeeKVzo_T688	0.02961	0.07061	0.0318	0.0803	0.01665	0.05938	0.3993	0.4186
interstitial_0_mp-75	0.03880	0.05961	0.0390	0.0634	0.00888	0.06206	0.0822	0.2177
interstitial_0_mp-75_T2750	0.02512	0.13636	0.0245	0.1423	0.01576	0.11345	0.1245	0.2788
interstitial_0_mp-75_T300	0.01292	0.11446	0.0153	0.1179	0.01189	0.09385	0.1385	0.1695
interstitial_0_mp-75_T688	0.03246	0.04785	0.0313	0.0521	0.00694	0.03807	0.1150	0.2048
md_0_mp-75	0.01577	0.00070	0.0172	0.0004	0.01298	0.00070	0.1187	0.0004
md_1_mp-109412	0.02966	0.12930	0.0315	0.1288	0.07591	0.15896	0.2196	0.2057
md_2_mp-8636	0.00633	0.00103	0.0090	0.0009	0.00690	0.00103	0.2354	0.0009
md_3_sc	0.06138	0.00022	0.0593	0.0003	0.03338	0.00022	0.6277	0.0003
md_4_InngSFTjbQPoKX5ZtDChEkeeKVzo	0.01471	0.00766	0.0149	0.0109	0.00611	0.01611	0.0972	0.0235
vacancy_0_mp-75	0.00581	0.05676	0.0036	0.0555	0.01103	0.03865	0.2017	0.1585
vacancy_0_mp-75_T2750	0.00383	0.11421	0.0046	0.1108	0.00692	0.09789	0.1918	0.2196
vacancy_0_mp-75_T300	0.00439	0.06385	0.0030	0.0656	0.00550	0.06065	0.1797	0.1672
vacancy_1_mp-109412	0.06134	0.10943	0.0587	0.1111	0.01561	0.11376	0.2964	0.2534
vacancy_1_mp-109412_T2750	0.01852	0.15383	0.0209	0.1577	0.00995	0.11849	0.1560	0.2760
vacancy_1_mp-109412_T300	0.03840	0.06731	0.0403	0.0678	0.02397	0.06322	0.1865	0.1063
vacancy_2_mp-8636	0.02474	0.04817	0.0212	0.0488	0.00655	0.04298	0.2657	0.1036
vacancy_2_mp-8636_T300	0.02260	0.07590	0.0193	0.0767	0.00589	0.06966	0.2554	0.1312
vacancy_2_mp-8636_T688	0.02601	0.04191	0.0227	0.0423	0.00803	0.03560	0.2694	0.0875
vacancy_3_sc	0.01681	0.09184	0.0221	0.0983	0.02469	0.05161	0.7688	0.0985
vacancy_3_sc_T2750	0.08667	0.15527	0.0846	0.1576	0.01083	0.23949	0.3591	0.3540
vacancy_3_sc_T300	0.04673	0.15189	0.0448	0.1502	0.00258	0.16785	0.3902	0.2988
vacancy_4_InngSFTjbQPoKX5ZtDChEkeeKVzo	0.00735	0.07235	0.0068	0.0725	0.00710	0.05322	0.1735	0.1212
vacancy_4_InngSFTjbQPoKX5ZtDChEkeeKVzo_T2750	0.02129	0.12703	0.0204	0.1256	0.01455	0.11559	0.2527	0.2487
vacancy_4_InngSFTjbQPoKX5ZtDChEkeeKVzo_T300	0.00667	0.16026	0.0051	0.1600	0.00821	0.14555	0.2478	0.2400
vacancy_4_InngSFTjbQPoKX5ZtDChEkeeKVzo_T688	0.03787	0.05858	0.0356	0.0586	0.00873	0.06261	0.2574	0.1038
EQGROUP	0.03345	0.08854	0.0343	0.0924	0.01479	0.08056	0.2912	0.2320
ALL	0.04423	0.09512	0.0467	0.1026	0.01581	0.08091	0.3414	0.3301

Table 16: Per-group full-data errors and held-out errors at 50% included (mean over ten draws) for Re.

Group	qSNAP				CLASP			
	δE_{full}	δF_{full}	δE_{50}	δF_{50}	δE_{full}	δF_{full}	δE_{50}	δF_{50}
elastic_0_mp-8_T3500	0.06409	0.54219	0.0614	0.5109	0.05234	0.50591	0.4729	0.8514
elastic_0_mp-8_T875	0.06917	0.42118	0.0673	0.3874	0.05029	0.42007	0.5444	0.8011
elastic_1_mp-1186901_T3500	0.06417	0.61518	0.0666	0.8274	0.04375	0.57085	3.4194	429.2231
elastic_1_mp-1186901_T875	0.05706	0.48272	0.0508	0.4557	0.04203	0.52838	0.6996	1.6970
elastic_3_mp-8642_T3500	0.08439	0.53392	0.0749	0.5002	0.04671	0.47160	0.4789	0.9749
elastic_3_mp-8642_T875	0.05310	0.38189	0.0489	0.3510	0.04056	0.38195	0.4796	0.8571
elastic_sc_T3500	0.12236	0.65116	0.1099	0.5863	0.07135	0.71890	0.5222	0.9213
elastic_sc_T875	0.06964	0.42519	0.0694	0.3809	0.08656	0.51384	0.5386	0.6438
interstitial_0_mp-8_T3500	0.06164	0.61904	0.0733	0.7954	0.03864	0.51646	0.2689	3.0674
interstitial_0_mp-8_T875	0.04780	0.43456	0.0449	0.4002	0.04169	0.42632	0.1929	0.6379
vacancy_0_mp-8_T3500	0.05068	0.58079	0.0440	0.5377	0.03349	0.50469	0.3491	0.7574
vacancy_0_mp-8_T875	0.09539	0.38460	0.0800	0.3708	0.01436	0.38451	0.3911	0.5275
vacancy_1_mp-1186901_T3500	0.05740	0.36970	0.0555	0.3449	0.04155	0.32737	0.3694	0.5033
vacancy_1_mp-1186901_T875	0.10078	0.29155	0.0906	0.2648	0.04342	0.25322	0.3945	0.3740
vacancy_2_mp-975065_T3500	0.03318	0.24156	0.0312	0.2325	0.04149	0.28519	0.4946	0.4474
vacancy_2_mp-975065_T875	0.04216	0.18957	0.0398	0.1798	0.01072	0.22569	0.4448	0.4237
vacancy_3_mp-8642_T3500	0.04510	0.55842	0.0394	0.5029	0.03570	0.49315	0.2930	0.7210
vacancy_3_mp-8642_T875	0.01586	0.29486	0.0219	0.2698	0.06151	0.32814	0.3712	0.4563
vacancy_sc_T3500	0.11044	0.70213	0.1000	0.6590	0.07343	0.67562	0.5896	0.9255
vacancy_sc_T875	0.20666	0.80438	0.1708	0.6639	0.12914	0.59804	0.6730	0.6904
EQGROUP	0.07255	0.47623	0.0670	0.4611	0.04994	0.45649	0.5994	22.2751
ALL	0.06779	0.49760	0.0652	0.5092	0.04927	0.47520	0.6348	44.5242

A.9 Per-group curves

Figures 6 and 7 give the tantalum data-efficiency curves group by group, for energies and for forces; Figs. 8, 9 and 10 give the first twenty-four groups of each transfer dataset. The high-symmetry tantalum groups (Volume_*, Elastic_*) are where the two descriptors are indistinguishable, and the low-symmetry ones are where they separate.

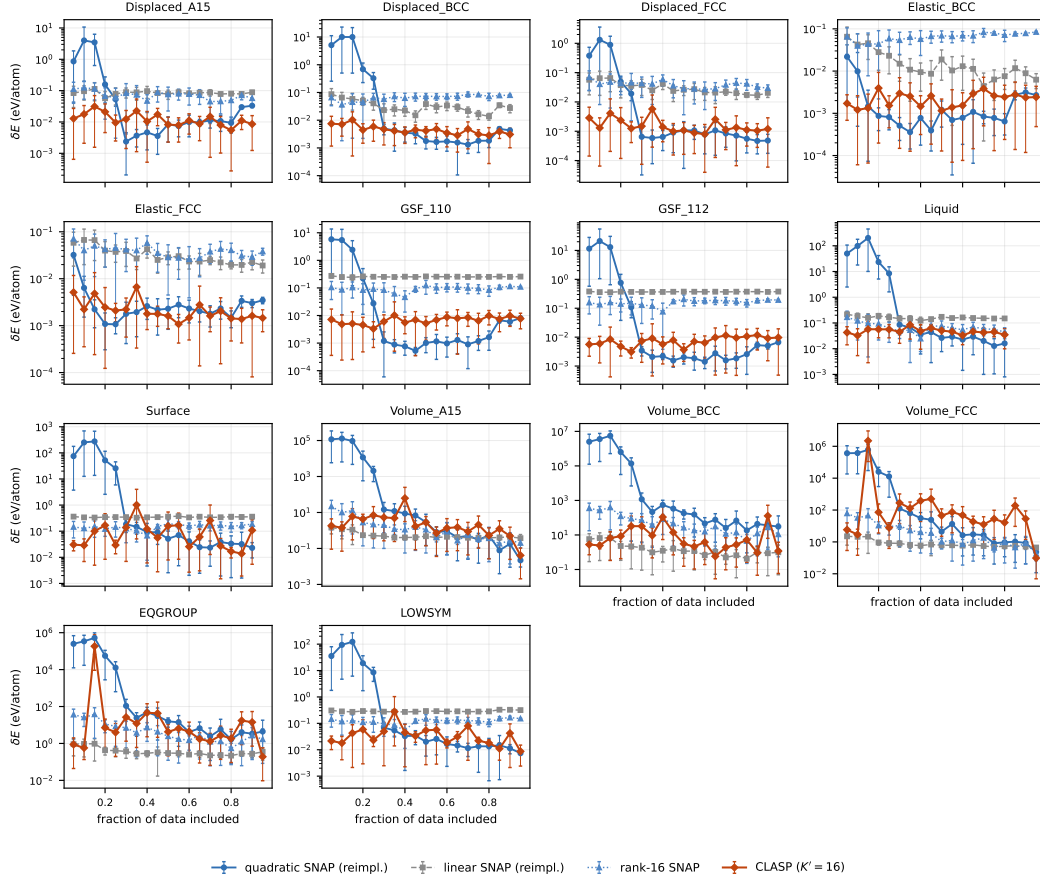


Figure 6: Tantalum: held-out energy error per training group, mean $\pm$ s.d. over the ten draws.

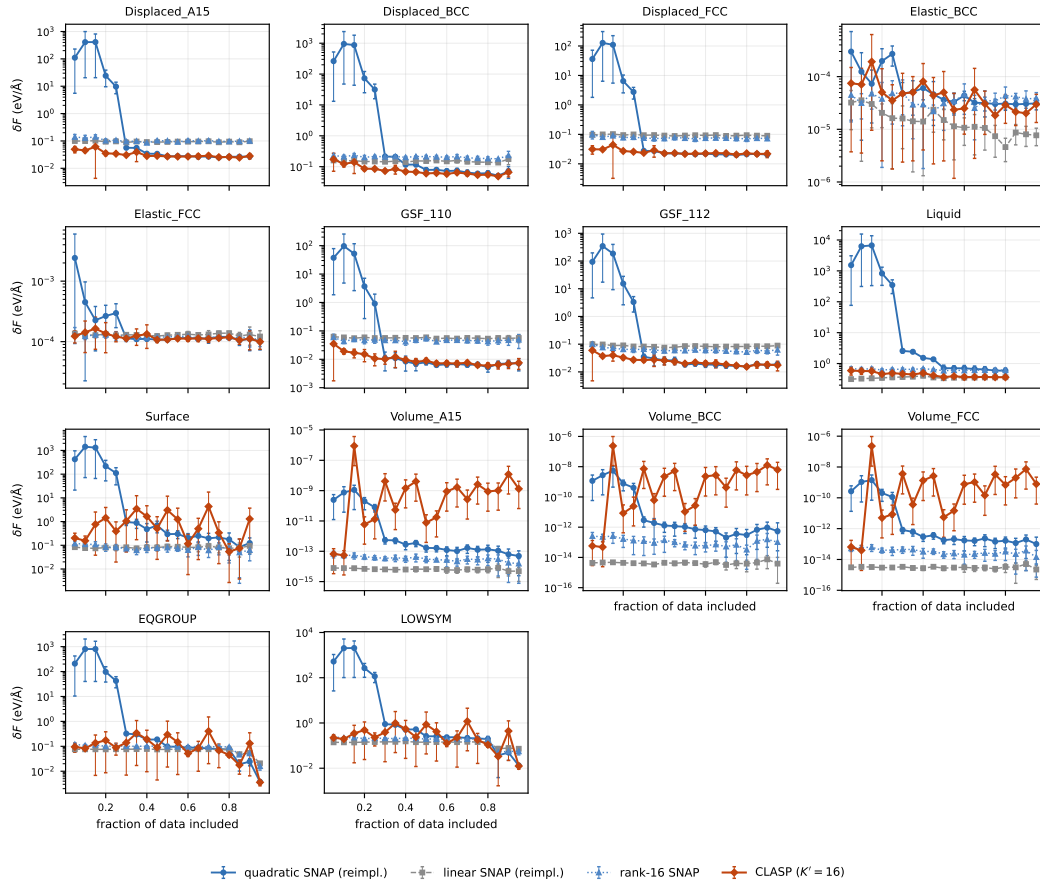


Figure 7: Tantalum: held-out force error per training group.

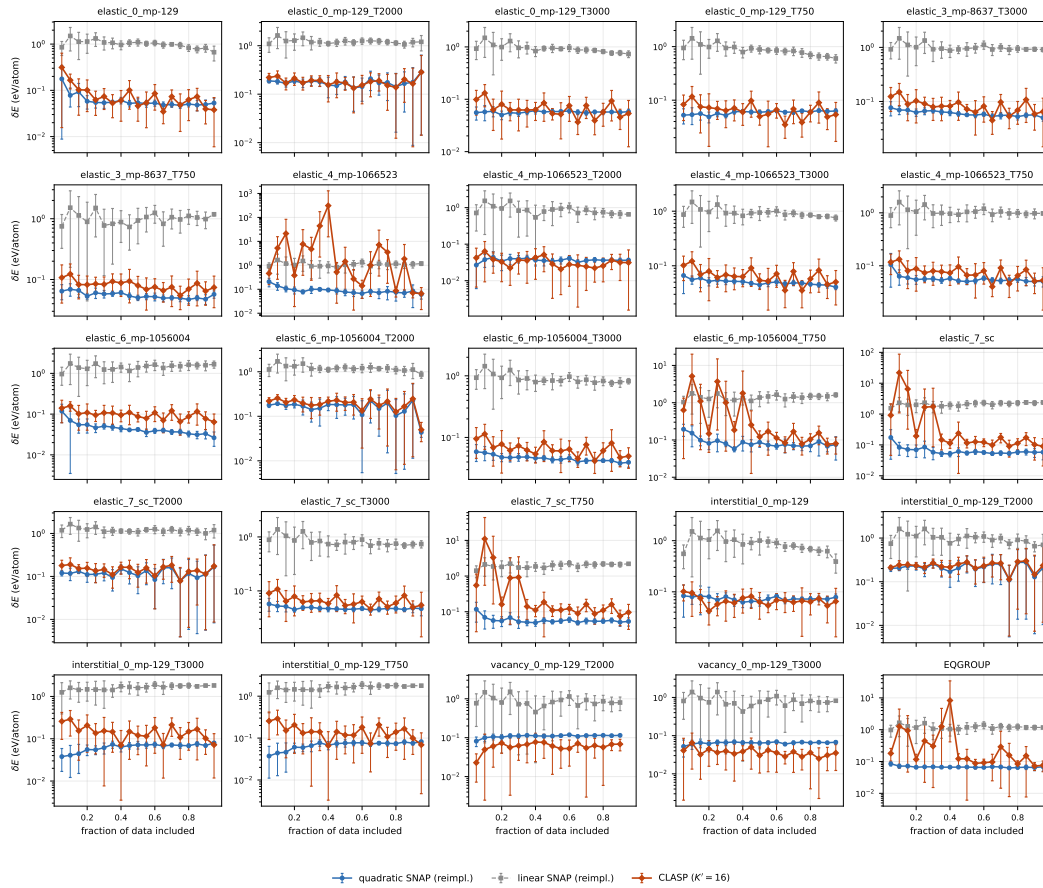


Figure 8: Molybdenum: held-out energy error for the first twenty-four prototype/defect/temperature groups.

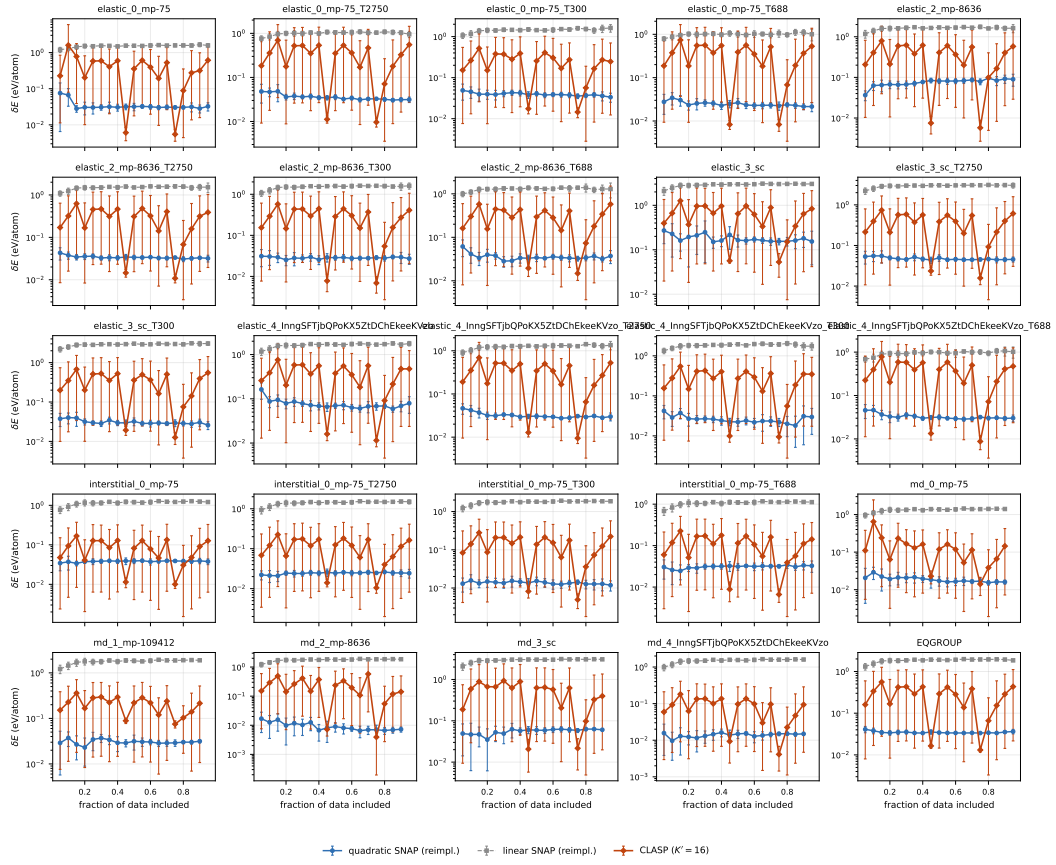


Figure 9: Niobium: held-out energy error, first twenty-four groups.

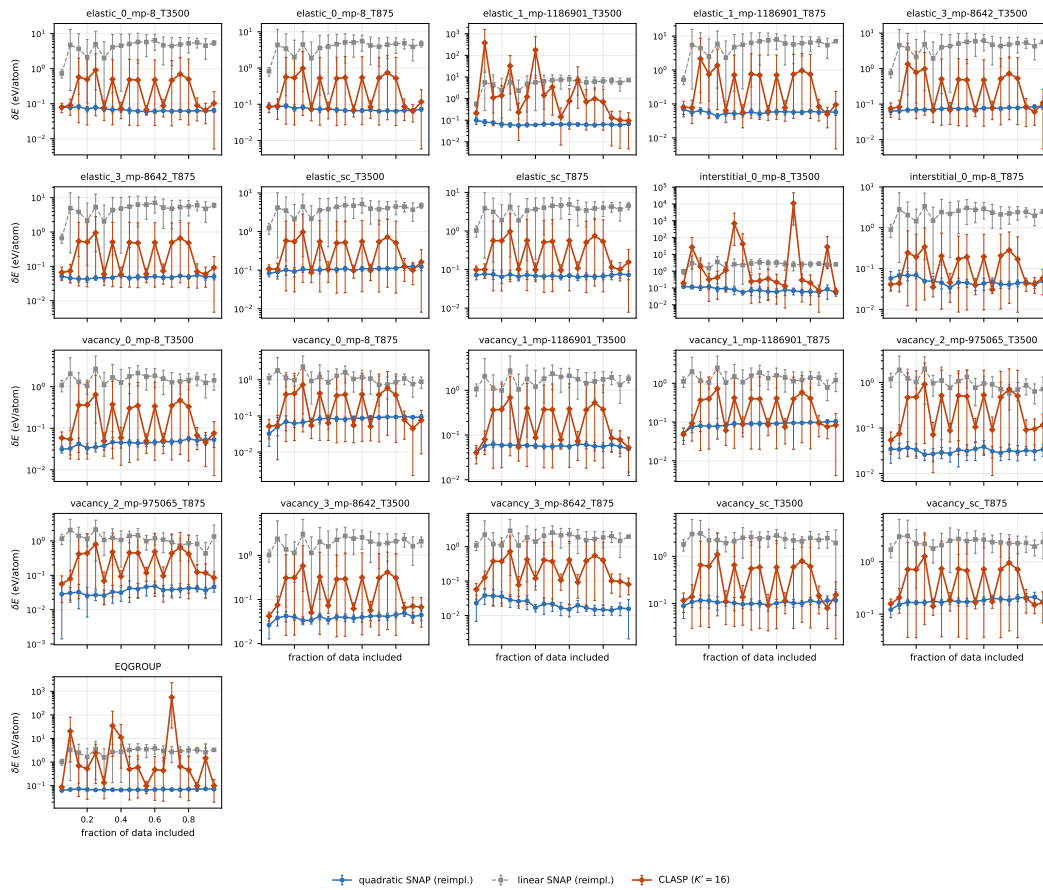


Figure 10: Rhenium: held-out energy error, all twenty groups.